Group Theory

Group Theory Notes

Useful Facts

- Let G be a finite group, and suppose G acts on itself by conjugation. Then, the stabilizer of $x \in G$ is denoted $C_G(x)$, and by the orbit-stabilizer theorem, it follows that the number of *elements* of the conjugacy class of x, $K_G(x)$, is given by $[G : C_G(x)]$. The *class equation* says we may find

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(x_i)],$$

  where $\{x_1, \dots, x_r\}$ is a family of representatives of conjugacy classes.
- If G acts on its *subgroups* by conjugation, then the *normalizer* of $H \leq G$, denoted $N_G(H)$, is the stabilizer of H under conjugation. Similarly, number of subgroups conjugate to $H \leq G$ is given by $[G : N_G(H)]$.
- If $P_1, \dots, P_r$ are p-Sylow subgroups of G, they are all conjugate to each other, and the normalizer $N_G(P_i)$ is a conjugate of $N_G\left(P_j\right)$ for $1 \leq i, j \leq r$.

Semidirect Products

Suppose N and H are subgroups of a group G, where N is normal and H is such that $N \cap H = \{e\}$. We will obtain a description of the structure of G in terms of the structure of N and H.

A short exact sequence of groups is a sequence of groups and homomorphisms

$$1 \to N \xrightarrow{\varphi} G \xrightarrow{\psi} \xrightarrow{H} \to 1$$

where ψ is surjective and φ identifies N with $\ker(\psi)$. In other words, the sequence is exact if N is normal in G and ψ induces an isomorphism $G/N \cong H$.

> **Definition:** Let N and H be groups. A group G is an *extension* of H by N if there is an exact sequence of groups
>
> $$1 \to N \to G \to H \to 1.$$
>
> The exact sequence is said to split if H can be identified with a subgroup of G, so that $N \cap H = \{e\}$.

If N is normal, then observe that any subgroup H of G acts on N by conjugation; furthermore, conjugation determines a homomorphism

$$\iota \colon H \to \operatorname{Aut}(N)$$

given by $h \mapsto \iota_h$, where $\iota_h(n) = hnh^{-1}$.

Now, we consider a certain tautology. If $G = NH$ with $N \cap H = \{e\}$ and N normal, then observe that

$$n_1 h_1 n_2 h_2 = \left(n_1\left(h_1 n_2 h_1^{-1}\right)\right)(h_1 h_2).$$

In particular, if we know the conjugation action of H on N, then we can recover the operation in G from this information and the operations in N and H.

This allows us to construct semidirect products by considering any two arbitrary groups N and H with an arbitrary group homomorphism

$$\begin{aligned} \theta \colon H &\to \operatorname{Aut}(N) \\ h &\mapsto \theta_h. \end{aligned}$$

Then, we may define an operation on $N \times H$ by taking

$$(n_1, h_1) \cdot (n_2, h_2) = \left(n_1 \theta_{h_1}(n_2), h_1 h_2\right).$$

Observe that in the special case where $N, H \leq G$ with $G = NH$, N normal, and $N \cap H = \{e\}$, the operation θ can be given by the map ι. The group $(N \times H, \cdot)$ is denoted $N \rtimes_\theta H$, and is called the semidirect product of N and H with respect to θ. Observe that the ordinary direct product is a semidirect product where θ is the trivial map.

We will discuss some examples.

(i) Suppose H is an abelian group of arbitrary order, and let $K = \mathbb{Z}/2\mathbb{Z}$. Define $\varphi \colon K \to \operatorname{Aut}(H)$ by inversion. Then, $G = H \rtimes K$ contains the subgroup H of index 2, with $xhx^{-1} = h^{-1}$ for all $h \in H$ and $e \neq x \in K$.

   When $H \cong \mathbb{Z}/n\mathbb{Z}$, this yields the dihedral group D_n of order $2n$, while if $H \cong \mathbb{Z}$, we have the infinite dihedral group D_∞.

(ii) If H is a group, and $K = \operatorname{Aut}(H)$, then $\varphi \colon K \to \operatorname{Aut}(H)$ given by the identity yields the semidirect product $H \rtimes \operatorname{Aut}(H)$, denoted the *holomorph* of H.

Solvable Groups

Definition: A series of subgroups G_i of a group G is a decreasing sequence of subgroups

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots,$$

where the length of the series is the number of strict inclusions.

A normal series is a series of subgroups where $G_{i+1} \trianglelefteq G_i$ for each i. The number $\ell(G)$ is the length of a maximal normal series.

A *composition series* is a normal series

$$G = G_0 \supsetneq G_1 \supsetneq \cdots \subsetneq G_n = \{e\}$$

where the successive quotients G_i/G_{i+1} are simple.

Theorem (Jordan–Hölder Theorem): Let G be a group, and suppose there are two composition series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}$$
$$G = G_0' \supsetneq G_1' \supsetneq G_2' \supsetneq \cdots \supsetneq G_m = \{e\}.$$

Then, $m = n$, and the quotient groups $H_i = G_i/G_{i+1}$ and $H_i' = G_i'/G_{i+1}'$ agree up to isomorphism and permutation.

Proof. Considering the following composition series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}$$

we induct on n. If $n = 0$, then G is trivial and there is nothing to prove.

Suppose $n > 0$, and consider another composition series

$$G = G_0' \supsetneq G_1' \supsetneq G_2' \supsetneq \cdots \supsetneq G_m' = \{e\}.$$

If $G_1 = G_1'$, then the result follows from the induction hypothesis since G_1 has a composition series of length $n-1$.

Therefore, we may assume that $G_1 \neq G_1'$. We observe that $G_1 G_1' = G$, since $G_1 G_1'$ is normal in G and $G_1 \supsetneq G_1 G_1'$, but there are no proper normal subgroups between G_1 and G since G/G_1 is simple.

Consider the group $K = G_1 \cap G_1'$. The distinct subgroups $G_i \cap K$ determine a composition series

$$K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}.$$

By the Second Isomorphism Theorem, we have

$$\begin{aligned}\frac{G_1}{K} &= \frac{G_1}{G_1 \cap G_1'}\\ &\cong \frac{G_1 G_1'}{G_1'}\\ &= \frac{G}{G_1'}\end{aligned}$$

and $G_1'/K \cong G/G_1$. These quotients are simple, so we obtain two new composition series

$$\begin{gathered}G \supsetneq G_1 \supsetneq K \supsetneq K_1 \supsetneq \cdots \supsetneq \{e\}\\ G \supsetneq G_1' \supsetneq K \supsetneq K_1 \supsetneq \cdots \supsetneq \{e\}.\end{gathered}$$

These composition series only differ at the first step. The two series have the same length and the same quotients. Next, we claim that the first of these series has the same length and quotients of the original series. Since

$$G_1 \supsetneq K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}$$

is a composition series for G_1, the induction hypothesis gives that it must have the same length and quotients as the composition series

$$G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\}.$$

Therefore, applying the induction hypothesis, it follows that the series

$$G_1' \supsetneq K \supsetneq K_1 \supsetneq K_2 \supsetneq \cdots \supsetneq K_r = \{e\}$$

has the same length and quotients as

$$G_1' \supsetneq G_2' \supsetneq \cdots \supsetneq G_m' = \{e\}.$$

□

Proposition: Let G be a group, and let N be a normal subgroup of G. Then, G has a composition series if and only if N and G/N have composition series. Furthermore,

$$\ell(G) = \ell(N) + \ell(G/N),$$

and the composition factors of G consist of the composition factors of N and of G/N.

Proof. If G/N has a composition series, then the subgroups appearing in it correspond to subgroups of G containing N with isomorphic quotients. Therefore, if both G/N and N have composition series, concatenating produces a composition series for G.

For the converse, suppose G has a composition series

$$G = G_0 \supsetneq G_1 \supsetneq G_2 \supsetneq \cdots \supsetneq G_n = \{e\},$$

and that N is a normal subgroup of G. Taking intersections with N gives a sequence of subgroups

$$N = G \cap N \supseteq G_1 \cap N \supsetneq \cdots \supsetneq \{e\} \cap N = \{e\}$$

such that $G_{i+1} \cap N$ is normal in $G_i \cap N$ for all i. We claim that this is a composition series upon deleting repetitions. For this, we will establish that the quotients $(G_i \cap N)/(G_{i+1} \cap N)$ are either trivial or isomorphic to G_i/G_{i+1}. Consider the homomorphism

$$G_i \cap N \hookrightarrow G_i \to \frac{G_i}{G_{i+1}}.$$

The kernel is $G_{i+1} \cap N$, so by the first isomorphism theorem, we have an injective homomorphism

$$\frac{G_i \cap N}{G_{i+1} \cap N} \hookrightarrow \frac{G_i}{G_{i+1}},$$

where we identify the image with a subgroup of G_i/G_{i+1}. This subgroup is normal since N is normal in G, and since G_i/G_{i+1} is simple, we get our desired result.

As for G/N, we have the composition series

$$\frac{G}{N} \supseteq \frac{G_1 N}{N} \supseteq \frac{G_2 N}{N} \supseteq \cdots \supseteq \{e_{G/N}\}.$$

such that $(G_{i+1}N)/N$ is normal in $(G_i N)/N$. By the third isomorphism theorem, the quotients

$$\frac{(G_i N)/N}{(G_{i+1}N)/N} \cong \frac{G_i N}{G_{i+1} N}$$

are isomorphic. Considering the homomorphism

$$G_i \hookrightarrow G_i N \to \frac{G_i N}{G_{i+1} N},$$

we have that this homomorphism is surjective, and the subgroup G_{i+1} of the source is sent to the identity of the target. By the first isomorphism theorem, there is a surjective homomorphism

$$\frac{G_i}{G_{i+1}} \to \frac{G_i N}{G_{i+1} N},$$

so that $G_i N/G_{i+1} N$ is either trivial or isomorphic to it as G_i/G_{i+1} is simple. $\square$

Composition series are the first major step in understanding the structure of groups. Next, we consider a special series known as the *derived series*.

Definition: If G is a group, then the commutator subgroup, denoted G' or $[G,G]$, is the subgroup generated by all the commutators $[g,h] = ghg^{-1}h^{-1}$.

The following follows directly from the definitions of the commutator subgroup.

Theorem: If G is a group, and G' is its commutator subgroup, then

(i) G' is normal in G (in fact, G' is *characteristic* in G);

(ii) the quotient G/G' is commutative;

(iii) G' yields the "maximal abelian quotient," in the sense that if $N \trianglelefteq G$ is another normal subgroup with G/N commutative, then $G' \leq G$.

Definition: The *derived series* is a descending sequence of subgroups

$$G \supseteq G' \supseteq G'' \supseteq G''' \supseteq \cdots.$$

Note that if G is commutative, then the sequence automatically hits $\{e\}$, and if G is simple and noncommutative, then $G = G' = G'' = \cdots$.

Definition: A group is called *solvable* if its derived series terminates in the identity.

We call a normal series abelian (resp. cyclic) if all the quotients are abelian (resp. cyclic).

Proposition: For a finite group G, the following are equivalent:

(i) all composition factors of G are cyclic;

(ii) G admits a cyclic series ending in $\{e\}$;

(iii) G admits an abelian series ending in $\{e\}$;

(iv) G is solvable.

Proof. Observe that (i) implies (ii) and (ii) implies (iii) follow immediately from the definition. Furthermore, (iii) implies (i) by refining an abelian series to a composition series into a family of cyclic p-groups. Furthermore, (iv) implies (iii) immediately from the fact that the derived series is an abelian series.

Therefore, we must show that (iii) implies (iv). Consider an abelian series

$$G = G_0 \supsetneq G_1 \supsetneq G_1 \supsetneq \cdots \supsetneq G_n = \{e\}.$$

Then, we claim that $G^{(i)} \subseteq G_i$, where $G^{(i)}$ denotes the ith iterated commutator subgroup.

This follows by induction. Since G/G_1 is commutative, it follows that $G' \subseteq G_1$. Since G_i/G_{i+1} is abelian, it follows that $G_i' \subseteq G_{i+1}$, so that

$$\begin{aligned} G^{(i+1)} &= \left(G^{(i)}\right)' \\ &\subseteq G_i' \\ &\subseteq G_{i+1}. \end{aligned}$$

Therefore, we have that $G^{(n)} \subseteq G_n = \{e\}$, so that the derived series terminates at $\{e\}$. $\square$

We will now discuss an application of solvability to understanding the structure of certain subgroups of the symmetric group. This has applications to Galois theory.

Proposition: Let p be prime. Then, any solvable subgroup of S_p with order divisible by p is contained in the normalizer of a p-Sylow subgroup of S_p. Furthermore, any such p-Sylow subgroup of S_p is a Frobenius group of order $p(p-1)$.

Proof. If G is a solvable subgroup of S_p with order divisible by p, then there exists a p-Sylow subgroup $P = \langle\sigma\rangle$ for some p-cycle σ. We will start by showing the second assertion.

If we identify the points that P acts on with $\mathbb{F}_p$, then we may take $\sigma = \left(x \overset{t_1}{\longmapsto} x+1\right)$. Observe that a permutation π that commutes with σ takes $\pi(x+1) = \pi(x)+1$; if $\pi(0) = c$, then $\pi = t_c$, meaning that $C_{S_p}(P) = P$, which has order p. There is thus an embedding $N_{S_p}(P)/P \hookrightarrow \operatorname{Aut}(\mathbb{Z}/p\mathbb{Z})$, which is isomorphic to $\mathbb{Z}/(p-1)\mathbb{Z}$. Observe that the dilations $x \overset{m_a}{\longmapsto} ax$ define every automorphism of $\mathbb{Z}/p\mathbb{Z}$, the automorphisms of P, $\sigma \mapsto \sigma^a$, are implemented by conjugation $m_a t_1 m_a^{-1} = t_a$. Thus, we have that

$$\begin{aligned} N_{S_p}(P) &= P \rtimes \langle\text{dilations}\rangle \\ &= \mathbb{Z}/p\mathbb{Z} \rtimes \mathbb{Z}/(p-1)\mathbb{Z}. \end{aligned}$$

Now, we will show that if G is a solvable subgroup of S_p, then there is exactly one p-Sylow subgroup. First, observe that in the derived series, the group $G^{(n-1)}$ has $\left[G^{(n-1)}, G^{(n-1)}\right] = D^{(n)}(G) = 1$, so that $G^{(n-1)}$ is abelian. Since every derived subgroup is characteristic, $G^{(n-1)}$ is normal in G.

Consider the stabilizer H of $0 \in \mathbb{F}_p$. The index of H is p, and observe that the group $G^{(n-1)}H$ is a subgroup of G that contains H, meaning that $\left[G : G^{(n-1)}H\right]$ divides p, so it is equal to either 1 or p. If $G^{(n-1)}H = H$, then $G^{(n-1)} \leq H$, so $G^{(n-1)}$ fixes 0, meaning $G^{(n-1)}$ fixes all of $\mathbb{F}_p$, implying that $G^{(n-1)}$ is contained in the kernel of a faithful action — specifically, the faithful action of S_p on $\mathbb{F}_p$ — so that $G^{(n-1)}H = G$.

By the second isomorphism theorem, we have

$$\begin{aligned} \left[G^{(n-1)} : G^{(n-1)} \cap H\right] &= \left[G^{(n-1)}H : H\right] \\ &= [G : H] \\ &= p, \end{aligned}$$

whence the size of a $G^{(n-1)}$-orbit of 0 has size p, so that $G^{(n-1)}$ is transitive.

Set $A_0 = G^{(n-1)} \cap H$. We will show that $A_0 = 1$, which occurs from showing that the intersection is normal in G and the fact that a normal subgroup fixing any point fixes all points. Observe that $G^{(n-1)}$ is abelian, so A_0 is normal in $G^{(n-1)}$. Furthermore, if $h \in H$ and $a \in A_0$, then $hah^{-1} \in G^{(n-1)}$ since $G^{(n-1)}$ is normal in G, and hah^{-1} fixes 0 since all three of h^{-1}, a, and h fix 0. Therefore, A_0 is normal in H. In particular, this means that A_0 is normal in both H and $G^{(n-1)}$, so A_0 is normal in $G = G^{(n-1)}H$. Yet, since $a \cdot 0 = 0$ for all $a \in A_0$, and the conjugates of a stabilizer are the stabilizers of the other elements, it follows that A_0 stabilizes every element of $\mathbb{F}_p$, so that A_0 is contained in the kernel of a faithful action, whence $A_0 = 1$.

Therefore, we have that $\left|G^{(n-1)}\right| = \left[G^{(n-1)} : A_0\right] = p$, whence $G^{(n-1)} \cong \mathbb{Z}/p\mathbb{Z}$, and since $G^{(n-1)}$ is a normal subgroup in G, it follows that there is a unique p-Sylow subgroup. □

Nilpotent Groups

Definition: Let G be a group. Define the following subgroups inductively:

$$\begin{aligned} Z_0(G) &= \{e\} \\ Z_1(G) &= Z(G) \\ Z_{i+1}(G)/Z_i(G) &= Z(G/Z_i(G)). \end{aligned}$$

That is, $Z_{i+1}(G)$ is the preimage of $Z(G/Z_i(G))$ under the canonical projection. The series of subgroups

$$Z_0(G) \leq Z_1(G) \leq Z_2(G) \leq \cdots$$

is called the *upper central series* of G.

A group G is called *nilpotent* if $Z_c(G) = G$ for some integer c. The smallest such c is called the *nilpotence class* of G.

Proposition: Let p be a prime, and let P be a group of order p^a. Then, P is nilpotent with nilpotence class at most $a-1$.

Proof. For each $i > 0$, we have that $P/Z_i(P)$ is a p-group, so we know that if $|P/Z_i(P)| > 1$, then $Z(P/Z_i(P)) \neq 1$ by the properties of p-groups.

If $Z_i(P) \neq P$, then $|Z_{i+1}(P)| \geq p|Z_i(P)|$, meaning that $|Z_{i+1}(P)| \geq p^{i+1}$. In particular, $|Z_a(P)| \geq p^a$, so $P = Z_a(P)$, so P is nilpotent of class at most a.

If P were of nilpotence class a, then this would hold if and only if $|Z_i(P)| = p^i$ for all i. Yet, this would give that $Z_{a-2}(P)$ would have index p^2 in P, so $P/Z_{a-2}(P)$ would be order p^2, hence abelian, meaning that $P/Z_{a-2}(P)$ would equal its center and $Z_{a-1}(P)$ would be equal to p. Therefore, P is nilpotent of class at most $a-1$. □

Proposition: The direct product of a finite number of nilpotent groups is nilpotent.

Proof. Suppose $G = H \times K$ (the rest follows inductively). Furthermore, inductively, we assume that $Z_i(G) = Z_i(H) \times Z_i(K)$, where the base case follows from the definition of the direct product. Let $\pi_H \colon H \to H/Z_i(H)$ be the canonical projection, and let π_K be defined analogously.

Then, we have that $\varphi \colon G \to G/Z_i(G)$ is given by

$$\begin{aligned} G &= H \times K \\ &\xrightarrow{\pi_H \times \pi_K} H/Z_i(H) \times K/Z_i(K) \\ &\xrightarrow{\psi} \frac{H \times K}{Z_i(H) \times Z_i(K)} \\ &= \frac{H \times K}{Z_i(H \times K)} \\ &= G/Z_i(G). \end{aligned}$$

For shorthand, write $\pi = \pi_H \times \pi_K$. Then, we have

$$\begin{aligned} Z_i(G) &= \varphi^{-1}(Z(G/Z_i(G))) \\ &= \pi^{-1} \circ \psi^{-1}(Z(G/Z_i(G))) \\ &= \pi^{-1}(Z(H/Z_i(H) \times K/Z_i(K))) \\ &= \pi^{-1}(Z(H/Z_i(H)) \times Z(K/Z_i(K))) \\ &= \pi_H^{-1}(Z(H/Z_i(H))) \times \pi_K^{-1}(Z(K/Z_i(K))) \\ &= Z_{i+1}(H) \times Z_{i+1}(K). \end{aligned}$$

Therefore, we have $Z_i(G) = Z_i(H) \times Z_i(K)$ for all i. Thus, G is nilpotent with nilpotence class equal to the larger of the nilpotence class of H and the nilpotence class of K. $\square$

Proposition: If H is a proper subgroup of a nilpotent group G, then H is a proper subgroup of its normalizer $N_G(H)$.

Proof. Let $Z_0(G) = \{e\}$, and let n be the largest index such that $Z_n(G) < H$; such an n exists since H is a proper subgroup and G is nilpotent. Select $a \in Z_{n+1}(G)$ with $a \notin H$. Then, for every $h \in H$, we have $ahZ_n(G) = (aZ_n(G))(hZ_n(G))$, and since $aZ_n(G)$ is in the center of $G/Z_n(G)$ by the definition of $Z_{n+1}(G)$, it follows that $ahZ_n(G) = haZ_n(G)$. In particular, we have $ah = h'ha$ for some $h' \in Z_n(G) < H$, whence $aha^{-1} \in H$ with $a \in N_G(H)$. Since $a \notin H$, it follows that H is a proper subgroup of $N_G(H)$. $\square$

Proposition: A finite group is nilpotent if and only if it is the direct product of its Sylow subgroups.

Proof. If G is a direct product of its p-Sylow subgroups, then G is nilpotent since it is the direct product of a family of nilpotent groups.

Now, suppose G is nilpotent and P is a p-Sylow subgroup. Either, $P = G$, in which case G is a p-group and we are done, or G is a proper subgroup of G. If P is a proper subgroup of G, then P is a proper subgroup of $N_G(P)$ by the previous proposition. Since the normalizer of $N_G(P)$ is itself, it follows that $N_G(P) = G$ since it cannot be a proper subgroup, so P is normal in G, hence it is the unique p-Sylow subgroup of G.

In the general case, let $|G| = p_1^{n_1} \cdots p_k^{n_k}$, where p_i are distinct primes and $n_i > 0$. Let $P_1, \dots, P_k$ be the corresponding proper normal Sylow subgroups corresponding to p_i. Since $|P_i| = p_i^{n_i}$ for each i, it follows that $P_1 P_2 \cdots \widehat{P_i} \cdots P_k$ is a subgroup in wihch every element has order dividing $|G|/p_i^{n_i}$.

Consequently, $P_i \cap \left(P_1 \cdots \widehat{P_i} \cdots P_k\right) = \{e\}$, so inductively, it follows that $P_1 \cdots P_k = P_1 \times \cdots \times P_k$. Therefore, since the order of G equals the order of the direct product, it follows that $G = P_1 \times \cdots \times P_k$. $\square$

We consider another, equivalent characterization of the nilpotent groups.

Definition: Let G be a group. Define the groups G_i inductively by $G_1 = G$ and $G_{i+1} = [G_i, G]$. The series

$$G = G_1 \geq G_2 \geq \cdots$$

defines the *lower central series* of G.

Lemma: Let $K \trianglelefteq G$ and $K \leq H \leq G$. Then, $[H, G] \leq K$ if and only if $H/K \leq Z(G/K)$.

Proof. If $h \in H$ and $g \in G$, then $hKgK = gKhK$ if and only if $[h,g]K = K$ if and only if $[h,g] \in K$. $\square$

In particular, this means that $G_i/G_{i+1} \leq Z(G/G_{i+1})$.

Theorem: Let G be a group. Then, G is nilpotent of nilpotence class c if and only if $G_{c+1} = \{e\}$. Moreover,

$$G_{i+1} \leq Z_{c-i}(G)$$

for all i.

Proof. Suppose $Z_c(G) = G$. We will prove that the inclusion holds by induction on i. If $i = 0$, then $G_1 = Z_c(G) = G$. Inductively, we have

$$G_{i+2} = [G_{i+1}, G]$$

$$\begin{aligned}&\le [Z_{c-i}(G), G]\\&\le Z_{c-i-1}(G),\end{aligned}$$

where the last inclusion emerges from the definition of $Z_k(G)$ and the previous lemma. Therefore, we have $G_{c+1} \le Z_0(G) = \{e\}$.

Now, suppose $G_{c+1} = \{e\}$. We will prove that $G_{c+1-j} \le Z_j(G)$ by induction on j. If $j = 0$, then $G_{c+1} = Z_0(G) = \{e\}$. By the third isomorphism theorem, we have a surjective homomorphism $G/G_{c+1-j} \to G/Z_j(G)$. Since $\left[G_{c-j}, G\right] = G_{c+1-j}$, we have that $G_{c-j}/G_{c+1-j} \le Z\left(G/G_{c+1-j}\right)$. Passing to the image of the homomorphism, we have $G_{c-j}Z_j(G)/Z_j(G) \le Z\left(G/Z_j(G)\right) = Z_{j+1}(G)/Z_j(G)$. Therefore, we have $G_{c-j} \le G_{c-j}Z_j(G) \le Z_{j+1}(G)$. □

A Simple Group of Order 168

We will work to understand the existence and uniqueness of a simple group of order $168 = 2^3 \cdot 3 \cdot 7$ by understanding the structure of its Sylow subgroups.

First, observe that $|G|$ does not divide $6!$, so G has no proper subgroup of index less than 7, as otherwise the action of G on the cosets of this subgroup would give an action into some S_n with $n \le 6$. Since G is simple, this action would necessarily be injective, implying that the order of G would divide the order of this S_n.

Next, since G is simple and the Sylow theorems give that $n_7 = 1$ or $n_7 = 8$, it follows that $n_7 = 8$. Since the number of Sylow subgroups is the index of the normalizer, this gives that the normalizer of a 7-Sylow subgroup is of order 21. Furthermore, no order 2 element normalizes a 7-Sylow subgroup, so G has no elements of order 14.

We will show now that G has no elements of order 21. Suppose x had order 21. Then, there is a 3-Sylow subgroup contained in $\langle x \rangle$, implying that the normalizer of this 3-Sylow subgroup would have order divisible by 7. Since we know that $n_3 | 8$, it follows that $n_3 = 4$ since $n_3 \equiv 1 \mod 3$, which would contradict that G has no subgroups of index less than 7. Therefore, $n_3 = 7$ or $n_3 = 28$.

Now, we will show that $n_3 = 28$. Suppose $n_3 = 7$. Let P be a 3-Sylow subgroup, and let T be a 2-Sylow subgroup of $N_G(P)$, where we note that $|N_G(P)| = 24$ (the order of T is 8). Since a 3-Sylow subgroup normalizes some 7-Sylow subgroup R of G, it follows that for every $t \in T$, since $tPt^{-1} = P$, we have P normalizes tRt^{-1}. Since T acts by conjugation on the eight 7-Sylow subgroups of G, and no element of order 2 in G normalizes a 7-Sylow subgroup, we must have that T acts transitively — that is, P normalizes all the 7-Sylow subgroups. Then, P is contained in the intersection of the normalizes of all the 7-Sylow subgroups; yet, since this intersection is a proper normal subgroup of G, it is trivial. Therefore $n_3 = 28$.

Next, we use a lemma.

Lemma: In a finite group G, if n_p is not congruent to 1 modulo p^k, then there are distinct p-Sylow subgroups P and R of G such that $P \cap R$ has index at most p^{k-1} in both P and R.

Proof. Let P act by conjugation on the set of all the p-Sylow subgroups. Suppose p^k divides $[P : P \cap R]$ for all p-Sylow subgroups R not equal to P. Then, each orbit has size divisible by p^k. Since the set of p-Sylow subgroups is equal to the sum of all the lengths of the orbits, then we have $n_p = 1 + \ell p^k$ for some integer ℓ. Thus, there is some R such that $[P : P \cap R] \le p^{k-1}$. □

Turning our attention to the 2-Sylow subgroups, we observe that $n_2 = 7$ or $n_2 = 21$. In particular, this means that n_2 is not congruent to 1 mod 8, so there are 2-Sylow subgroups T_1 and T_2 with nontrivial intersection such that $U = T_1 \cap T_2$ is of maximum order.

Now, we claim that U is a Klein 4-group with $N_G(U) \cong S_4$. Define $N = N_G(U)$. We have that $|U| = 2$ or $|U| = 4$, and N permutes the non-identity elements of U by conjugation. Observe that a subgroup of order 7 in N would then commute with some element of order 2 in U, implying that an element of order 2 would normalize a 7-Sylow subgroup, which cannot happen. Therefore, the order of N is not divisible by 7.

Remark: In the setting of the lemma, we observe that if $P_0 = P \cap Q$ is an intersection of p-Sylow subgroups, then $P_0 < P$ and $P_0 < R$, so since p-groups are nilpotent, we have $P_0 < N_P(P_0)$ and $P_0 < N_Q(P_0)$. In the special case where $R = P \cap Q$ is maximal, then $R < N_P(R) \leq N_G(R)$ and $R < N_Q(R) \leq N_G(R)$.

Now, consider a p-Sylow subgroup H of $N_G(R)$. Then, there is a p-Sylow subgroup S of G such that $R \leq H \leq S$. Suppose it were the case that there were separate p-Sylow subgroups of $N_G(R)$ H_1 and H_2 that were both contained in S. Then, since $H_i \leq N_G(R)$, it follows that $H_i \leq N_S(R)$. Notice that $N_S(R)$ is a p-group that is a subgroup of $N_G(R)$; by maximality, it follows that $N_S(R) \leq H_i$ as the H_i are p-Sylow subgroups, so that $N_S(R) = H_1 = H_2$, a contradiction. Therefore, if $H_1 \leq S_1$ and $H_2 \leq S_2$ are distinct p-Sylow subgroups of G that contain $N_G(R)$, we have that $R \leq H_1 \cap H_2 \leq S_1 \cap S_2$. Yet, by maximality, it follows that $S_1 \cap S_2 = R$.

Suppose now toward contradiction that $N_G(R)$ has a unique p-Sylow subgroup H. Notice that $N_P(R) \leq H$ and $N_Q(R) \leq H$, and there is some p-Sylow subgroup $S$of G such that $H \leq S$. This gives that $R < N_P(R) \leq P \cap S$ and $R < N_Q(R) \leq Q \cap S$. If (without loss of generality) $S = P$, then $N_Q(R) \leq S$ and $N_Q(R) \leq Q$, so $N_Q(R) \leq P \cap Q = R$, yet $R < N_P(R)$, a contradiction. If S is not equal to S or P, it follows that P and S are distinct p-Sylow subgroups of G with $N_G(R) \leq P \cap S$, whence $|P \cap S| > |R|$.

In particular, this means that if $P \cap Q$ is maximal, we have that $N_G(P \cap Q)$ has more than one p-Sylow subgroup.

In the case of the group $N = N_G(U)$, we have automatically that N has more than one 2-Sylow subgroup, so $|N| = 2^a \cdot 3$ with $a = 2$ or $a = 3$. Suppose P is a 3-Sylow subgroup of N. Since P is a 3-Sylow subgroup of G, it follows that the group $N_N(P)$ has order 3 or 6 (since the order of $N_G(P)$ is 6). Therefore, by the Sylow theorems, N must have 4 3-Sylow subgroups that are permuted transitively by conjugation. Observe that any group of order 12 must have either a normal 2-Sylow subgroup or a normal 3-Sylow subgroup, but since N has more than one 2-Sylow subgroup, it follows that $|N| = 24$.

Let K be the kernel of N acting by conjugation on its 4 3-Sylow subgroups. Then, K is the intersection of the normalizers of the 3-Sylow subgroups of N. If $K = \{e\}$, then $N \cong S_4$. Else, suppose that $K \neq \{e\}$. Since we have that $K \leq N_N(P)$, it follows that K has order dividing 6, and since P does not normalize any other 3-Sylow subgroup, we cannot have $P \leq K$, so $|K| = 2$. Yet, N/K is then a group of order 12 that has more than one 2-Sylow subgroup and has 4 3-Sylow subgroups. Therefore, $N \cong S_4$. Now, we know that S_4 has a unique nontrivial normal 2-subgroup, which is the Klein 4-group, giving that U is a Klein 4-group.

Since $N \cong S_4$, it follows that N contains a 2-Sylow subgroup of G, and that $N_N(P) \cong S_3$, so that $N_G(P) \cong S_3$ and the 2-Sylow subgroups of G are isomorphic to the dihedral group D_4 of order 8. Furthermore, G has no elements of order 6.

We thus find that any element of order 2 does not commute with an element of order either 3 (else we would have an element of order 6) or 7 (since the order of the normalizers of the 7-Sylow subgroups is 21). If T is a 2-Sylow subgroup, then $T \cong D_4$, so $Z(T) \cong \langle z \rangle$ for some element z of order 2. In particular, this means $T \leq C_G(z)$; since $|C_G(z)|$ has no odd prime factors, it follows that $C_G(z) = T$. Furthermore, any element normalizing T normalizes its center, hence commutes with z, meaning that the 2-Sylow subgroups of G are self-normalizing, so that $n_2 = 21$.

Now, since $|C_G(z)| = 8$, the element z has 21 conjugates. Additionally, G has one conjugacy class of elements of order 4 by the fact that the 2-Sylow subgroups are isomorphic to D_4 (so the elements of order 4 are conjugate to each other in their subgroup), and this conjugacy class has 42 elements. Furthermore, there are 48 elements of order 7 and 56 elements of order 3. Summing these gives all 167 non-identity elements. In particular, this means that all the elements of order 2 are conjugate to each other.

Let T be a 2-Sylow subgroup, and let $U \leq T$ be the aforementioned Klein 4-group in T, and let W be the other Klein 4-group. From the Sylow theorems, since U and W are not conjugate in the normalizer $N_G(T) = T$, it follows that U and W are not conjugate in G.[I] Next, we claim that $N_G(W) \cong S_4$. For this, suppose we have $W = \langle z, w \rangle$ where $Z(T) = \langle z \rangle$. Since w is conjugate in G to z (both have order 2), it follows that $C_G(w) = T_0$ is another 2-Sylow subgroup of G containing W different from T. In particular, this

[I] This follows from from "Frattini's Argument," which says that if H is a normal subgroup of G and P is a p-Sylow subgroup of H, then $N_G(P)H = G$.

means that $W = T \cap T_0$, so since U is an arbitrary maximal intersection of 2-Sylow subgroups of G, the same argument that gives $N_G(U) \cong S_4$ also gives $N_G(W) \cong S_4$.

All this analysis gives the following.

Proposition: If G is a simple group of order 168, then the following hold:

(i) $n_2 = 21$, $n_3 = 7$, and $n_7 = 8$;

(ii) the 2-Sylow subgroups of G are isomorphic to the dihedral group D_4, while the 3 and 7-Sylow subgroups are cyclic;

(iii) G is isomorphic to a subgroup of A_7, and G has no subgroup of index strictly less than 7;

(iv) the following are the conjugacy classes of G

- $\{e\}$;
- 2 classes of elements of order 7 with 24 elements per class;
- 1 class of elements of order 3 with 56 elements;
- 1 class of elements of order 2 containing 21 elements;

(v) if T is a 2-Sylow subgroup, and U and W are the Klein 4-subgroups of T, then U and W are not conjugate in G, and have $N_G(U) \cong N_G(W) \cong S_4$;

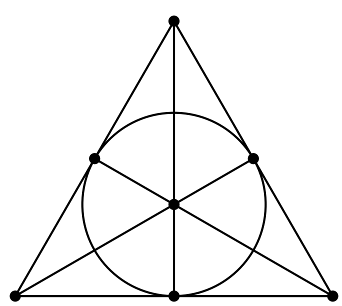
(vi) G has 3 conjugacy classes of maximal subgroups, two of which are isomorphic to S_4 and one of which is isomorphic to the non-abelian group $\mathbb{Z}/7\mathbb{Z} \rtimes \mathbb{Z}/3\mathbb{Z}$ of order 21.

However, none of this shows existence. To find this existence, we will construct a special geometric object that emerges from the action of G on its subgroup structure.

Let $U_1, \dots, U_7$ be the conjugates of U, and let $W_1, \dots, W_7$ be the conjugates of W. We will call the U_i points, and the W_j lines. Next, we will define an incidence relation by saying that U_i is on line W_j if and only if U_i normalizes W_j. Equivalently, U_i is on the line W_j if and only if $U_i W_j \cong D_8$,[II] which occurs if and only if W_j normalizes U_i.

Inside each point (or line) stabilizer (isomorphic to S_4) there is a unique normal 4-subgroup V, and three other non-normal 4-subgroups which we will refer to as B_1, B_2, and B_3. The groups VB_i are the 3 2-Sylow subgroups of S_4. Therefore, we have that each line contains exactly 3 points and every point lies on exactly 3 lines. Since any two non-normal 4-subgroups of S_4 generate S_4 (and determine the other two Klein 4-subgroups of that copy of S_4), it follows that any two points on a line uniquely determine the line. Finally, by counting, every pair of points lies on a unique line, and each pair of lines intersects at a unique point.

This geometry is known as the *projective plane of order* 2, or the *Fano Plane*, denoted $\mathcal{F}$, and is depicted below.



Now, we observe that every $g \in G$ acts by conjugation on the set of points and lines, and preserves the inci-

[II] There are two Klein 4-groups in the normalizer $N_G(W_j)$, and since U_i is not conjugate to W_j, it follows that U_i normalizes W_j if and only if U_i is the other Klein 4-group in $N_G(W_j)$.

dence relation, with only the identity element in G that fixes all the points. Therefore, G is isomorphic to a subgroup of $\text{Aut}(\mathcal{F})$. Any automorphism of $\mathcal{F}$ that fixes any two points on a line and fixes a third point not on the line will necessarily fix all points, meaning that any automorphism of $\mathcal{F}$ is determined by its action on any three non-collinear points. There are 168 such triples, so $\mathcal{F}$ has at most 168 automorphisms, so that if the simple group G exists, then it is unique and isomorphic to $\text{Aut}(\mathcal{F})$.

To determine the automorphisms of $\mathcal{F}$, we consider a different construction. Let $V = \mathbb{F}_2^3$ be the 3-dimensional vector space over $\mathbb{F}_2$. Call the 7 1-dimensional subspaces of V the points and the 7 2-dimensional subspaces the lines. Say a point is incident to a line via subspace inclusion. Observe that $\text{GL}(V)$ acts faithfully on the points and lines and preserves incidence, so $\text{Aut}(\mathcal{F})$ has order at least 168, so $\text{Aut}(\mathcal{F}) \cong \text{GL}(V) = \text{GL}_3(\mathbb{F}_2)$, with $\text{Aut}(\mathcal{F})$ having order 168.

Finally, we will prove that $\text{GL}(V)$ is simple. Suppose H is a proper nontrivial normal subgroup of $\text{GL}(V)$. Suppose Ω is the set of points, and N is the stabilizer in $\text{GL}(V)$ of a point in Ω. The intersection of all the conjugates of N fixes all points, so this intersection is the identity. Therefore, $H \not\leq N$, and we have $\text{GL}(V) = HN$.

By the second isomorphism theorem, $[H : H \cap N] = [HN : N]$, so 7 divides $|H|$. Furthermore, since $\text{GL}(V)$ is isomorphic to a subgroup of S_7 and the normalizers of 7-Sylow subgroups of S_7 have order 42, it follows that $\text{GL}(V)$ does not have a normal 7-Sylow subgroup, so $n_7(\text{GL}(V)) = 8$.

A normal 7-Sylow subgroup of H is characteristic in H, hence normal in $\text{GL}(V)$, meaning that H does not have a unique 7-Sylow subgroup. Since $n_7(H) \equiv 1 \bmod 7$, and $n_7(H) \leq n_7(\text{GL}(V)) = 8$, it follows that $n_7(H) = 8$. Thus, $|H|$ is divisible by 8, so 56 divides H, and since H is proper, $|H| = 56$. By applying a Sylow-style counting argument to H, we see that H has a normal (hence characteristic) 2-Sylow subgroup, which is thus normal in $\text{GL}(V)$. Yet, this would mean that $\text{GL}(V)$ has a unique 2-Sylow subgroup. Note that the upper and lower triangular matrices are subgroups of $\text{GL}_3(\mathbb{F}_2)$ with order 8, so we achieve a contradiction.

Group Theory Worked Solutions

Problem (August '05, Problem 2): Let G be a finitely generated group.

(a) If H is a finite group, show that there exist only finitely many group homomorphisms from G to H.

(b) If n is a given natural number, show that there exist only finitely many normal subgroups n of G with $[G : N] = n$.

Solution:

(a) Since H is finite, it is finitely generated and finitely presented, the latter of which follows from the fact that a finite group is fully determined by its Cayley table. We observe that any homomorphism $G \to H$ must map generators of G to generators of H and satisfy the relations of both G and H; since there are finitely many generators and relations for H, it follows that there are finitely many such homomorphisms.

(b) We observe that any normal subgroup N of G arises as the kernel of the homomorphism $G \to G/N$, so in particular, $[G : N] = n$ if and only if N is the kernel of a homomorphism from G to a group of order n. There are finitely many such groups since any group of order n must embed in S_n by Cayley's theorem, and there are finitely many subgroups of S_n of any order, so by part (a), it follows that there are finitely many group homomorphisms from G to any group of order n, so there are finitely many normal subgroups N with $[G : N] = n$.

Problem (August '11, Problem 7): Let G be a finite simple group of order 168.

(a) How many elements of order 7 does G have? Why?

(b) How may conjugacy classes of elements of order 7 does G have?

(c) Exhibit two elements of $G = \text{GL}_3(\mathbb{F}_2)$ of order 7 that are not conjugate in G. Explain why they are not conjugate.

Solution:

(i) We start by considering the factorization $168 = 2^3 \cdot 3 \cdot 7$. The number of 7-Sylow subgroups is congruent to 1 mod 7 and divides 24. This gives the possible values of either 1 or 8. Since G is simple, there cannot be only one 7-Sylow subgroup, so there are 8 7-Sylow subgroups. These 7-Sylow subgroups must intersect at the identity, so there are 48 elements of order 7.

(ii) Since there are 8 7-Sylow subgroups, it follows that the normalizer of any such 7-Sylow subgroup has order 21. In particular, this means that any 3-Sylow subgroup must normalize a 7-Sylow subgroup, and that no 2-Sylow subgroup normalizes a 7-Sylow subgroup. Suppose toward contradiction that there were an element of order 21. Then, it would follow that the normalizer of any 3-Sylow subgroup would have order divisible by 7, meaning that the number of 3-Sylow subgroups would divide 8. This means there would be exactly 4 3-Sylow subgroups. Acting G on the set of these subgroups by conjugation yields a homomorphism $G \to S_4$, which by the Sylow theorems is nontrivial. Yet, since G is simple, this means that the homomorphism is injective, implying that $|G| \mid |S_4|$, which is a contradiction. Therefore, there cannot be any element of order 21.

Given $x \in P$, where P is a 7-Sylow subgroup, observe that $P \leq C_G(x) \leq N_G(P)$. We have that the order of $C_G(x)$ is either 7 or 21. Suppose that the order of $C_G(x)$ is 21. By Cauchy's Theorem, there is an element g of order 3 which generates a subgroup of order 3. Yet, this means that gx has order 21, which contradicts what we have just shown. Therefore, the order of $C_G(x)$ is 7. The number of elements in the conjugacy class is thus $[G : C_G(x)] = 24$, so that there are 2 conjugacy classes of elements of order 7.

Problem (August '06, Problem 3): Show that any nilpotent group G of order 900 is abelian.

Solution: We know that any nilpotent group is equal to the direct product of its Sylow subgroups. Observe that $900 = 2^2 \cdot 3^2 \cdot 5^2$, so if we let P_2, P_3, P_5 denote the respective Sylow subgroups, we have that

$$G \cong P_2 \times P_3 \times P_5.$$

Notice that each of these groups is of order p_i^2, meaning that for each of these groups, we have $|Z(P_i)| \geq p_i$. Yet, if $|Z(P_i)| = p$ (i.e., assuming that the P_i are not abelian), then $|P_i/Z(P_i)| = p_i$, meaning that $P_i/Z(P_i)$ would be cyclic, hence P_i would be abelian, which is a contradiction. In particular, this means that the P_i are abelian. Since G is the direct product of abelian groups, it is abelian.

Problem (August '16, Problem 4): Fix a group G.

(a) Show that if N is a normal p-Sylow subgroup of G and H is a subgroup of order not divisible by p, then HN is a subgroup of G isomorphic to a semidirect product $N \rtimes H$.

(b) Consider a group G of order 255. Show that G is cyclic.

Solution:

(a) We start by observing that if h is a non-identity element in H, then h cannot have an order divisible by p, and we observe that any element of N has order of the form p^k. In particular, this means that $N \cap H = \{e\}$ in G. Since N is normal in G, it follows that $HN = NH$, so that HN is a subgroup of G. Furthermore, $|HN| = |H||N|$, so we have all the necessary requirements for HN to be an internal semidirect product isomorphic to $N \rtimes H$.

(b) We start by seeing that G is a group of order $3 \cdot 5 \cdot 17$. Observe that there is a unique 17-Sylow subgroup since the number of 17-Sylow subgroups divides 15 and is congruent to 1 mod 17. If P is this subgroup, then $|G/P| = 15$. Since any group of order 15 is necessarily cyclic since there is no nontrivial homomorphism from $\mathbb{Z}/3\mathbb{Z}$ to $\text{Aut}(\mathbb{Z}/5\mathbb{Z})$ or vice versa, it follows that $G \cong \mathbb{Z}/15\mathbb{Z} \times \mathbb{Z}/17\mathbb{Z}$, so G is cyclic.

Problem (August '19, Problem 1):

(a) Let G be a group of order 12. Prove that G has a normal 2-Sylow subgroup or a normal 3-Sylow subgroup.

(b) Prove that there are at least 5 pairwise non-isomorphic groups of order 12.

Solution:

(a) Observe that we may factor $12 = 2^2 \cdot 3$. Therefore, there exists a 2-Sylow subgroup and a 3-Sylow subgroup. The number of 3-Sylow subgroups is congruent to 1 mod 3 and divides 4, so it is equal to either 1 or 4. If there is one 3-Sylow subgroup, then this subgroup is normal and we are done. Else, if there are 4, these subgroups intersect at the identity, so that there are 8 elements of order 3. This gives 4 elements (including the identity) whose order does not divide 3, meaning that this forms our 2-Sylow subgroup, which is necessarily normal.

(b) We start by considering the abelian groups of order 12. By the classification theorem, it follows that there are two such groups

$$\begin{aligned} G_1 &= \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \\ G_2 &= \mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}. \end{aligned}$$

These groups are pairwise non-isomorphic since G_1 does not have any elements of order 4.

Now, we start by considering non-abelian groups. We start by considering the map $\mathbb{Z}/2\mathbb{Z} \to \operatorname{Aut}(\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z})$ given by $0 \mapsto \operatorname{id}$, $1 \mapsto \left(w \mapsto w^{-1}\right)$. This yields a semidirect product $\mathbb{Z}/2\mathbb{Z} \rtimes \mathbb{Z}/6\mathbb{Z} = D_6$.

Next, we observe that the alternating group A_4 is of order 12. We note that A_4 is not isomorphic to D_6 since D_6 has an element of order 6 while there are no elements of order 6 in A_4.

Finally, since $\mathbb{Z}/4\mathbb{Z}$ and $\operatorname{Aut}(\mathbb{Z}/3\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ are of the same order, there is a nontrivial homomorphism $\mathbb{Z}/4\mathbb{Z} \to \operatorname{Aut}(\mathbb{Z}/3\mathbb{Z})$ giving rise to a semidirect product $\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$. This group is not isomorphic to either D_6 or A_4 since neither D_6 nor A_4 contain an element of order 4.

Problem (DF 6.2.15)**:** Classify groups of order 105.

Solution: We may factor $105 = 3 \cdot 5 \cdot 7$. First, we show that there is a normal subgroup of either order 5 or order 7. For this, observe that the number of 5-Sylow subgroups is either 1 or 21, while the number of 7-Sylow subgroups is either 1 or 15. If there is one of either, then we are done, while if there are 21 5-Sylow subgroups and 15 7-Sylow subgroups, we get 84 elements of order 5 and 90 elements of order 7, which implies that there are at least 174 elements in G, a contradiction.

We start by showing that the 7-Sylow subgroup is normal. Suppose there are 15 7-Sylow subgroups, yielding 90 elements of order 7. Then, there is one 5-Sylow subgroup, yielding 4 elements of order 5. This gives 10 elements of order 3, implying that there are 5 3-Sylow subgroups. Yet, since the number of 3-Sylow subgroups must be congruent to 1 mod 3, this yields a contradiction. Therefore, there is exactly 1 7-Sylow subgroup.

If P_7 is this 7-Sylow subgroup, and P_5 is a 5-Sylow subgroup, then $H = P_5P_7$ is a group of order 35, which is necessarily cyclic since there is no nontrivial homomorphism $\mathbb{Z}/5\mathbb{Z} \to \operatorname{Aut}(\mathbb{Z}/7\mathbb{Z})$. Notice that $[G : H] = 3$, so since the index is given by the smallest prime that divides G, it follows that H is in fact normal. Furthermore, this means that P_5 is normal, so there is a unique 5-Sylow subgroup as well.

There is a homomorphism $\mathbb{Z}/3\mathbb{Z} \to (\mathbb{Z}/35\mathbb{Z})$ given by $1 \mapsto [11]$, which yields a semidirect product $(\mathbb{Z}/35\mathbb{Z}) \rtimes \mathbb{Z}/3\mathbb{Z}$.

Ring Theory

Ring Theory Notes

Localization

If R is a (commutative, unital) ring, then we may consider a subset $S \subseteq R$ such that $1 \in S$, $0 \notin S$, and for any $s_1, s_2 \in S$, we have $s_1s_2 \in S$ (such an S is called a *multiplicative set*). Then, we may construct the *ring of fractions* $S^{-1}R$ as the set of equivalence classes in $R \times S$ with $(r_1, s_1) \sim (r_2, s_2)$ if and only if $r_1s_2 - r_2s_1 = 0$. We write $[(r,s)] = \frac{r}{s}$, and define addition and multiplication by

$$\begin{aligned}\frac{r_1}{s_1} + \frac{r_2}{s_2} &= \frac{r_1 s_2 + r_2 s_1}{s_1 s_2}\\ \frac{r_1}{s_1}\frac{r_2}{s_2} &= \frac{r_1 r_2}{s_1 s_2}.\end{aligned}$$

There is a natural embedding ι of R into $S^{-1}R$ by taking $r \mapsto \frac{r}{1}$.

Theorem: Let R be a commutative ring with unity, S a multiplicatively closed subset, and $\phi\colon R \to T$ a ring homomorphism that maps S to units in T. Then, there is a unique homomorphism $\varphi\colon S^{-1}R \to T$ such that $\varphi \circ \iota = \phi$.

$$\begin{tikzcd} R \arrow[r, "\iota"] \arrow[dr, "\phi"'] & S^{-1}R \arrow[d, "\varphi"] \\ & T \end{tikzcd}$$

Proof. Define

$$\varphi\left(\frac{r}{s}\right) = \phi(r)\phi(s)^{-1}.$$

□

Primality in the Gaussian Integers

We will discuss primality in the Gaussian integers. Recall that the Gaussian integers are given by

$$\begin{aligned}\mathbb{Z}[i] &= \{a + bi \mid a, b \in \mathbb{Z}\}\\ &\cong \frac{\mathbb{Z}[x]}{(x^2+1)}.\end{aligned}$$

The Gaussian integers are equipped with a norm $N : \mathbb{Z}[i] \to \mathbb{Z}_{\geq 0}$ given by $N(a+bi) = (a+bi)(a-bi) = a^2+b^2$, which is a multiplicative norm satisfying $N(zw) = N(z)N(w)$. Furthermore, with this norm, the Gaussian integers form a Euclidean domain. The units of $\mathbb{Z}[i]$ are ± 1 and $\pm i$.

Lemma: Let $q \in \mathbb{Z}[i]$. Then, there is a prime integer $p \in \mathbb{Z}$ such that $N(q) = p$ or $N(q) = p^2$.

Proof. Since q is not a unit, $N(q) \neq 1$. Therefore, $N(q)$ is a nontrivial product of prime integers, and since q is prime in $\mathbb{Z}[i]$, q must divide one of the prime integer factors of $N(q)$. Let p be this prime integer.

Then, $q|p$ in $\mathbb{Z}[i]$, so $N(q)|N(p) = p^2$, so $N(q) = p$ or $N(q) = p^2$. □

Example: We observe that 3 is a prime element of $\mathbb{Z}[i]$. Since $\mathbb{Z}[i]$ is a Euclidean domain, it is a UFD, so irreducibles are prime, meaning we only need to show that 3 is irreducible in $\mathbb{Z}[i]$.

Since $N(3) = 9$, it follows that the norm of any factor of 3 must be a divisor of 9. Therefore, it follows that the norm of any divisor of 3 is equal to 1, 3, or 9. If the norm is 9, then the factor is an associate of 3, while if the norm is 1, then the factor is a unit, so it follows that any nontrivial factor of 3 would have norm equal to 3. Yet, there are no such elements.

Meanwhile, 5 is not a prime element of $\mathbb{Z}[i]$, since we may write $5 = (2+i)(2-i)$.

We will now focus on proving one of the most celebrated theorems in number theory: Fermat's Theorem on sums of squares. We will say a prime integer $p \in \mathbb{Z}$ *splits* in $\mathbb{Z}[i]$ if it is not a prime element of $\mathbb{Z}[i]$. For example, 5 splits in $\mathbb{Z}[i]$ while 3 does not.

Lemma: A positive prime integer $p \in \mathbb{Z}$ splits in $\mathbb{Z}[i]$ if and only if it is the sum of two squares in $\mathbb{Z}$.

Proof. Suppose $p = a^2 + b^2$ with $a, b \in \mathbb{Z}$. Then,

$$p = (a+bi)(a-bi)$$

in $\mathbb{Z}[i]$, so since $N(a \pm bi) = a^2 + b^2 = p \neq 1$, it follows that neither of the factors is a unit in $\mathbb{Z}[i]$, so that p is not irreducible, hence not prime.

Conversely, suppose p is not irreducible in $\mathbb{Z}[i]$. Then, p has an irreducible factor $q \in \mathbb{Z}[i]$ that is not an associate of p. Since $q|p$, multiplicativity of the norm gives that $N(q)|N(p) = p^2$, hence $N(q) = p$ as q and p are not associates and q is not a unit. If we have $q = a + bi$, then

$$\begin{aligned} p &= N(q) \\ &= a^2 + b^2. \end{aligned}$$

□

Next, we discuss a characterization of splitting via a single congruence.

Lemma: A positive odd prime integer p splits in $\mathbb{Z}[i]$ if and only if it is congruent to 1 modulo 4.

Proof. Our goal is to understand when $\mathbb{Z}[i]/(p)$ is an integral domain. Yet, we have the series of isomorphisms

$$\begin{aligned} \frac{\mathbb{Z}[i]}{(p)} &\cong \frac{\mathbb{Z}[x]/\left(x^2+1\right)}{(p)} \\ &\cong \frac{\mathbb{Z}[x]}{(p, x^2+1)} \\ &\cong \frac{\mathbb{Z}[x]/(p)}{(x^2+1)} \\ &\cong \frac{\mathbb{Z}/p\mathbb{Z}[x]}{(x^2+1)}. \end{aligned}$$

In particular, this means that p splits in $\mathbb{Z}[i]$ if and only if $x^2 + 1$ has a root in $\mathbb{Z}/p\mathbb{Z}$, or that there is some n such that $n^2 \equiv -1$ modulo p. We will show that this condition is equivalent to p being congruent to 1 modulo 4.

Let $G = (\mathbb{Z}/p\mathbb{Z})^\times$, and fix a generator g. Observe that p is odd, so $p - 1 = 2\ell$ for some integer ℓ. Let $\overline{n}$ denote the class of an integer n mod (p). For any $n \notin (p)$, it follows that there is some m such that $\overline{n} = g^m$.

Observe that the class $\overline{-1}$ generates the unique subgroup of order 2 of G, and since g^ℓ does the same, it follows that $g^\ell = -1$. In particular, we have that $n^2 = -1$ modulo p if and only if $g^{2m} = g^\ell$, or that $2m \equiv \ell$ mod 2ℓ, or that $(p-1) = 2\ell$ is a multiple of 4, or $p \equiv 1$ mod 4. □

This gives the following.

Theorem (Fermat's Theorem on Sums of Squares)**:** A positive odd prime $p \in \mathbb{Z}$ is a sum of two squares if and only if $p \equiv 1$ mod 4.

Ring Theory Worked Solutions

Problem (August '05, Problem 3)**:** Let R be a commutative ring with 1 such that any ideal in R is principal. Let $S \subseteq R$ be a multiplicatively closed subset of R. Show that any ideal in the ring of fractions $S^{-1}R$ is also principal.

Solution: We claim that any ideal in $S^{-1}R$ is of the form $S^{-1}J$ for some ideal J of R. For this, consider a proper ideal I (one may choose their favorite maximal ideal) of $S^{-1}R$, and consider the ideal $J = I \cap R$. We claim that $J \cap S = \emptyset$.

Suppose not, and let $u \in J \cap S$, so by definition of the copy of R in $S^{-1}R$, we have $\frac{u}{1} \in J \cap S$. Since $J \cap S$ is an ideal, it follows that $\frac{u}{1}\frac{1}{u} \in J \cap S$, so $1 \in J \cap S$, so $1 \in J$, meaning $1 \in I$, meaning I is not proper, which is a contradiction.

Since $J \cap S = \emptyset$, it follows that $S^{-1}J$ is a well-defined set. We start by showing that $I \subseteq S^{-1}J$. For this, consider an element $\frac{a}{b} \in I$. Then, we have $\frac{a}{1} = \frac{a}{b}\frac{b}{1} \in I$, so $\frac{a}{b} \in S^{-1}J$. Additionally, if we have some element

$\frac{a}{b} \in S^{-1}J$, then we have $\frac{a}{b} = \frac{1}{b}\left(\frac{a}{1}\right)$, where by definition, $\frac{a}{1} \in I$ as $\frac{a}{1} \in I \cap R$, so that $\frac{a}{b} \in I$ since I absorbs multiplication.

In the special case where every ideal of R is principal, it follows that every ideal of $S^{-1}R$ is of the form $S^{-1}I$, meaning that every ideal of $S^{-1}R$ is principal generated by $\left(\frac{a}{1}\right)$, where $I = (a)$.

Problem (August '20, Problem 4)**:** Let R be a commutative ring with 1. An ideal I of R is called irreducible if I cannot be written as $I = J \cap K$, where J and K are both ideals strictly containing I.

(a) Prove that every prime ideal is irreducible.

(b) Assume that R contains a nonzero nilpotent element. Prove that R contains an irreducible ideal which is not prime.

Solution:

(a) Let P be a prime ideal of R. Suppose toward contradiction that there exist proper ideals J and K with $P = J \cap K$ and $P \subsetneq J, K$. Choose $a \in J \setminus P$ and $b \in K \setminus P$. Then, $ab \in P$ since $ab \in J \cap K$ by absorption of multiplication. Yet, by the definition of prime ideal, it follows that either $a \in P$ or $b \in P$, which contradicts the definition of a and b.

(b) Suppose $a \in R$ is a nonzero nilpotent element. We consider the set $\mathcal{S}$ of ideals that do not contain a, ordered by inclusion. This set is nonzero since the zero ideal is in this set and a is not zero. If $\mathcal{C}$ is a chain in $\mathcal{S}$, then the union of the ideals in $\mathcal{C}$ is an upper bound in $\mathcal{S}$. If it were not the case that the union was in $\mathcal{S}$, then we would have some ideal $I \in \mathcal{C}$ with $a \in I$, which would imply that $I \notin \mathcal{S}$.

Therefore, by Zorn's Lemma, there is a maximal element M of $\mathcal{S}$. Observe that the nilradical of R is the intersection of all the prime ideals, so since M does not contain a, it follows that M is not prime.

Suppose toward contradiction that M is not irreducible. Then, there would be ideals J and K where $M \subsetneq J, K$ with $M = J \cap K$. Without loss of generality, if $a \notin J$, then it would follow that $J \in \mathcal{S}$, which would violate the maximality of M. Meanwhile, if $a \in J$ and $a \in K$, then we would have $a \in J \cap K = M$, which would violate the fact that $M \in \mathcal{S}$.

Problem (January '22, Problem 3)**:** Let $f(x,y), g(x,y) \in \mathbb{C}[x,y]$ be two polynomials that do not have a non-constant common factor.

(i) Show that f and g are relatively prime as elements of $\mathbb{C}(x)[y]$ and $\mathbb{C}(y)[x]$, where $\mathbb{C}(x)$ and $\mathbb{C}(y)$ are the fields of rational functions (i.e., the fraction fields of $\mathbb{C}[x]$ and $\mathbb{C}[y]$ respectively). Give a detailed argument.

(ii) Show that the system of polynomial equations $f(x,y) = 0$ and $g(x,y) = 0$ has finitely many solutions in $\mathbb{C}^2$.

Solution:

(i) Considering $\mathbb{C}[x,y] \cong (\mathbb{C}[x])[y]$ as the ring of polynomials in y over the integral domain $\mathbb{C}[x]$, then since $\mathbb{C}$ is a field, $\mathbb{C}[x]$ is a PID, hence a UFD, so $\mathbb{C}[x,y]$ is a UFD. In particular, the factorizations of $f(x,y)$ and $g(x,y)$ into irreducible polynomials in y over the underlying domain $\mathbb{C}[x]$ is unique. By Gauss's Lemma, irreducible polynomials in y over $\mathbb{C}[x]$ remain irreducible over $\mathbb{C}(x)$, so $f(x,y)$ and $g(x,y)$ do not share any factors in $\mathbb{C}(x)[y]$, hence they are relatively prime. A similar argument applies to $\mathbb{C}(y)[x]$ by switching the roles of x and y.

(ii)

Problem (January '19, Problem 3)**:** Let $R = \mathbb{Z}[x]$. Find the number of maximal ideals of R which contain $x^2 + 1$ and 15, and find explicit generators for each such ideal.

Solution: If J is a maximal ideal containing $x^2 + 1$ and 15, then J contains the ideal $\left(x^2 + 1, 15\right)$ generated by the two elements. By the Fourth Isomorphism Theorem, the maximal ideals containing $\left(x^2 + 1, 15\right)$ correspond to the maximal ideals in the quotient

$$S = \frac{\mathbb{Z}[x]}{(x^2+1,15)},$$

which by the Third Isomorphism Theorem, is isomorphic to

$$\begin{aligned} S &= \frac{\frac{\mathbb{Z}[x]}{(x^2+1)}}{\frac{(x^2+1,15)}{(x^2+1)}} \\ &\cong \frac{\mathbb{Z}[i]}{(15)}. \end{aligned}$$

The maximal ideals in S thus correspond to maximal ideals in $\mathbb{Z}[i]$ that contain 15. Since $\mathbb{Z}[i]$ is a Euclidean domain, it is a PID, hence a UFD, so irreducible elements are prime, and prime ideals are maximal, so equivalently, we are finding the irreducible elements that divide 15. By the classification of irreducibles in $\mathbb{Z}[i]$, we have that 3 is irreducible, so we must find the irreducibles that divide 5. We claim that $2+i$ and $2-i$ are irreducible in $\mathbb{Z}[i]$.

For this, observe that the norm on $\mathbb{Z}[i]$ given by $N(a+bi) = a^2+b^2$ is multiplicative, and that $N(2 \pm i) = 5$. Therefore, any irreducible that divides $2 \pm i$ must have norm equal to either 1 or 5. Since any element with norm 1 is a unit (hence not an irreducible) it follows that there are no irreducible elements of $\mathbb{Z}[i]$ that are not associated to $2 \pm i$ (respectively). Therefore, it follows that there are three maximal ideals of R that contain both x^2+1 and 15.

Problem (August '02, Problem 5): Let R be a commutative ring, $\mathfrak{a}_1, \dots, \mathfrak{a}_n$ ideals of R. If, for all $i \neq j$, we have $\mathfrak{a}_i + \mathfrak{a}_j == R$, then $\mathfrak{a}_1 \cap \cdots \cap \mathfrak{a}_n = \mathfrak{a}_1 \cdots \mathfrak{a}_n$.

Solution: We start with the case of $n = 2$. That is, we will show that if $\mathfrak{a}_1 + \mathfrak{a}_2 = R$, then $\mathfrak{a}_1 \cap \mathfrak{a}_2 = \mathfrak{a}_1\mathfrak{a}_2$. We already know that $\mathfrak{a}_1 \cap \mathfrak{a}_2 \supseteq \mathfrak{a}_1\mathfrak{a}_2$, so we will show that if $r \in \mathfrak{a}_1 \cap \mathfrak{a}_2$, then $r \in \mathfrak{a}_1\mathfrak{a}_2$. For this, observe that

$$r = r \cdot 1,$$

and since $\mathfrak{a}_1 + \mathfrak{a}_2 = R$, it follows that there are $a_1 \in \mathfrak{a}_1$ and $a_2 \in \mathfrak{a}_2$ such that $a_1 + a_2 = 1$. Therefore, we have

$$\begin{aligned} r &= r \cdot 1 \\ &= r(a_1 + a_2) \\ &= ra_1 + ra_2 \\ &\in \mathfrak{a}_1\mathfrak{a}_2. \end{aligned}$$

Problem (August '21, Problem 4): Let $R = \mathbb{Z}\left[\sqrt{-11}\right]$, and let $I = \left(3, 1+\sqrt{-11}\right)$ be the ideal of R generated by 3 and $1+\sqrt{-11}$.

(i) Prove that I is maximal.

(ii) Prove that I is not principal.

Solution:

(i) Via an evaluation homomorphism taking $p(x) \mapsto p\left(\sqrt{-11}\right)$, we observe that the map $\mathbb{Z}[x] \to \mathbb{Z}\left[\sqrt{-11}\right]$ yields an isomorphism

$$\frac{\mathbb{Z}[x]}{(x^2+11)} \cong \mathbb{Z}\left[\sqrt{-11}\right].$$

We call the left-hand side R'. Pulling back the ideal $\left(3, 1+\sqrt{-11}\right)$ into R' yields the ideal

$$I' = \frac{(3, 1+x) + \left(x^2+11\right)}{(x^2+11)}$$

in R'. Therefore, by the third isomorphism theorem, we have

$$\begin{aligned}
\mathbb{Z}\left[\sqrt{-11}\right]/\left(3, 1+\sqrt{-11}\right) &\cong R'/I' \\
&\cong \frac{\frac{\mathbb{Z}[x]}{\left(x^2+11\right)}}{\frac{(3,1+x)+\left(x^2+11\right)}{\left(x^2+11\right)}} \\
&\cong \frac{\mathbb{Z}[x]}{(3, 1+x)} \\
&\cong \frac{\frac{\mathbb{Z}[x]}{(3)}}{\frac{(3,1+x)}{(3)}} \\
&\cong \frac{(\mathbb{Z}/3\mathbb{Z})[x]}{(1+x)} \\
&\cong \mathbb{Z}/3\mathbb{Z},
\end{aligned}$$

where the final isomorphism emerges from the fact that $(x+1)$ is the kernel of the evaluation map $p(x) \mapsto p(-1)$. Therefore, the ideal I is maximal since the quotient by I yields a field.

(ii) Define a norm $N\colon \mathbb{Z}\left[\sqrt{-11}\right] \to \mathbb{N}$ given by

$$N\left(a + b\sqrt{-11}\right) = a^2 + 11b^2.$$

Notice that this norm is the restriction of the traditional norm $|z|^2 = z\overline{z}$ on $\mathbb{C}$, so in particular it is multiplicative. Suppose toward contradiction that there exists $q = a+b\sqrt{-11}$ such that $\left(3, 1+\sqrt{-11}\right) = (q)$. By the definition of the ideal (q), we have that $q|3$ and $q|1+\sqrt{-11}$, so that $N(q)|9$ and $N(q)|12$. Yet, this means that $N(q)|3$, so either $N(q) = 1$ or $N(q) = 3$. If $N(q) = 1$, then $q = 1$, implying that the ideal is not proper, which is a contradiction to what we have shown in (i). Therefore, $N(q) = 3$. Yet, there do not exist $a, b \in \mathbb{Z}$ such that $a^2 + 11b^2 = 3$, so we reach a contradiction.

Field/Galois Theory

Field Theory Notes

Irreducibility of Cyclotomic Polynomials

We will let Φ_n denote the nth cyclotomic polynomial, defined by

$$\Phi_n(x) = \prod_{\gcd(d,n)=1} \left(x - \zeta_n^d\right).$$

Recall that the cyclotomic polynomials satisfy

$$x^n - 1 = \prod_{d|n} \Phi_d(x).$$

Theorem: If p does not divide n, then the irreducible factors of $\overline{\Phi_n(x)} \in \mathbb{F}_p[x]$ are distinct and each has degree equal to the order of p modulo n.

Proof. We observe that $\Phi_n(x)|(x^n-1)$ in $\mathbb{Z}[x]$, so the divisibility relation is preserved when reducing modulo p, meaning that $\overline{\Phi_n(x)}$ is separable in $\mathbb{F}_p[x]$ as $x^n-\overline{1}$ is separable in $\mathbb{F}_p[x]$ by the assumption that $\gcd(p,n)=1$.

Let α be a root of $\overline{\Phi_n(x)}$ in an extension of $\mathbb{F}_p$. We will show that α is a primitive nth root of unity. Since $\overline{\Phi_n(x)}|\left(x^n-\overline{1}\right)$, it follows that we have $\alpha^n=1$. If α were not of order n, then it has some order m that properly divides n. In particular, this means that α is a root of some $\Phi_d(x)$ with $d|n$ properly. In particular, since $d|n$, it follows that $\overline{\Phi_n(x)\Phi_d(x)}$ divides $x^n-\overline{1}$, meaning that α is a double root of $x^n-\overline{1}$, contradicting the fact that $x^n-\overline{1}$ does not have any repeated roots. Therefore, α is a primitive root of unity.

If $\pi(x)$ is an irreducible factor of $\overline{\Phi_n(x)}$ in $\mathbb{F}_p[x]$, and α is a root of $\pi(x)$, then α is a primitive nth root of unity, so the degree of π, which is equal to $\left[\mathbb{F}_p(\alpha):\mathbb{F}_p\right]$, is the order of p modulo n. □

Field Theory Worked Solutions

Problem (September '84, Problem 7)**:** Factor x^8-1 into irreducibles in $\mathbb{Z}/p\mathbb{Z}[x]$ for all primes p.

Solution: We start by understanding the case where $p=2$. Then, we observe that $x^8-1=x^{2^3}-1$, so that $x^8-1=(x-1)^{2^3}$.

We start by observing that over $\mathbb{Z}$, x^8-1 factors into irreducibles given by

$$x^8-1=\Phi_1(x)\Phi_2(x)\Phi_4(x)\Phi_8(x),$$

where $\Phi_n(x)$ denotes the nth cyclotomic polynomial. In particular, we observe that these cyclotomic polynomials are given by

$$\begin{aligned}\Phi_1(x)&=x-1\\ \Phi_2(x)&=x+1\\ \Phi_4(x)&=x^2+1\\ \Phi_8(x)&=x^4+1.\end{aligned}$$

Therefore, we must understand the factorization into irreducibles over $\mathbb{Z}/p\mathbb{Z}$ of x^2+1 and x^4+1. When p is congruent to 1 modulo 4, then -1 is a square modulo p, so we may factor $x^2+1=(x-a)(x+a)$. When p is not congruent to 1 modulo 4, then -1 is not a square modulo p, meaning that x^2+1 is irreducible.

Now, we observe that the square of every odd number is congruent to 1 modulo 8, so that x^8-1 divides $x^{p^2-1}-1$, so we have that x^4+1 factors completely over $\mathbb{F}_{p^2}$. In particular, we may factor x^4+1 into the product of the minimal polynomials of the roots of x^4+1 in $\mathbb{F}_{p^2}$ over $\mathbb{F}_p$.

Problem (August '99, Problem 9)**:** If p is a prime number congruent to 1 mod 8, show that 2 is a quadratic residue mod p; i.e., there is an integer a such that $a^2\equiv 2$ modulo p.

Solution: First, we show that there exists some α such that $\alpha^4=-1$. For this, it suffices to show that the polynomial x^4+1 is not irreducible over $\mathbb{F}_p$. Since $\mathbb{F}_p$ is equal to the roots of the polynomial $p(x)=x^p-x$, we observe then that since $8|p-1$, it follows that x^8-1 divides $x^p-x=x\left(x^{p-1}-1\right)$. Therefore, we have $x^4+1|p(x)$. Since $p(x)$ splits over $\mathbb{F}_p$, it follows that x^4+1 splits over $\mathbb{F}_p$, so that x^4+1 has a root in $\mathbb{F}_p$.

Consider now the element $a=\alpha+\alpha^{-1}$. Then, we have

$$\begin{aligned}a^2&=\left(\alpha+\alpha^{-1}\right)^2\\ &=\alpha^2+\alpha^{-2}+2\\ &=2\end{aligned}$$

mod p, since $\alpha^{-2}=-\alpha^2$.

Problem (August '21, Problem 7):

(a) Let $n \in \mathbb{N}$, and let F be an algebraically closed field of characteristic 0 or of characteristic $p > 0$ where $p \nmid n$. Prove that F contains a primitive nth root of unity.

(b) According to (a), $\overline{\mathbb{F}_3}$, the algebraic closure of the field of three elements contains a 13th root of unity. Call it ω. Compute $[\mathbb{F}_3(\omega) : \mathbb{F}_3]$.

Solution:

(a) If the characteristic of F equals zero, the smallest subfield of F containing 1 is isomorphic to $\mathbb{Q}$, so we will assume without loss of generality that $\mathbb{Q} \subseteq F$. The nth cyclotomic polynomial, $\Phi_n(x)$, then has $\operatorname{spl}_{\mathbb{Q}}(\Phi_n(x)) \subseteq F$ is a separable extension that contains a primitive nth root of unity. Therefore, F has a primitive nth root of unity.

If the characteristic of F equals p, then we observe that the polynomial $p(x) = x^n - 1$ over $\mathbb{F}_p$ has $p'(x) = nx^{n-1} \neq 0$, whence $p(x)$ splits fully into linear factors in the algebraic closure F as $p(x)$ is then separable. There is then some extension $\mathbb{F}_{p^d}$ of $\mathbb{F}_p$ containing all the roots of $p(x)$, and in particular, we have $n | d-1$. Since any element of $\left(\mathbb{F}_{p^d}\right)^{\times}$ is a root of unity, it follows that there is some primitive $(d-1)$th root of unity contained in $\mathbb{F}_{p^d}$, whence there is also a primitive nth root of unity.

(b) If ω is a primitive 13th root of unity, then ω is a root of the polynomial $p(x) = x^{13} - 1$. We observe that over $\mathbb{Z}$, hence over $\mathbb{F}_3$, we have that $p(x) | x^{26} - 1$, whence we have $p(x)$ divides $q(x) = x^{27} - x$. In particular, this means that $p(x)$ splits completely in $\mathbb{F}_{p^3} \subseteq F$ and does not split over $q_2(x) = x^9 - x$, so that $[\mathbb{F}_3(\omega) : \mathbb{F}_3] = 3$.

Problem (August '99, Problem 8): If F is finite, show that every element $\alpha \in F$ is the sum $\alpha = \beta_1^2 + \beta_2^2$.

Solution: Since F is finite, we have that $F = \mathbb{F}_{p^n}$ for some prime p and some $n \geq 1$.

If $p = 2$, then the Frobenius map is given by squaring, and since F is finite, the map is surjective, so for any $\alpha \in \mathbb{F}_{2^n}$, there is some $\beta \in \mathbb{F}_{2^n}$ such that $\alpha = \beta^2$.

If $p \neq 2$, we consider the sets

$$\begin{aligned} A &= \left\{\beta^2 \,\middle|\, \beta \in F\right\} \\ B &= \{\alpha - \zeta \mid \zeta \in A\}. \end{aligned}$$

Observe that $\left(\mathbb{F}_{p^n}\right)^{\times}$ is cyclic, so $|A| = \frac{p^n+1}{2}$, and similarly, we have $|B| = \frac{p^n+1}{2}$. Yet, this means that $A \cap B \neq \emptyset$, so that there are some $\beta_1^2, \beta_2^2 \in A$ such that $\alpha - \beta_1^2 = \beta_2^2$, whence $\alpha = \beta_1^2 + \beta_2^2$.

Problem (August '08, Problem 3): Prove or disprove the following statements about irreducibility.

(a) If $\operatorname{Gal}(E/\mathbb{Q}) = S_n$ for E the splitting field over $\mathbb{Q}$ of a polynomial $f(x) \in \mathbb{Q}[x]$ of degree n, then $f(x)$ must be irreducible.

(b) If α is algebraic over $\mathbb{Q}$, then its minimal polynomial $\mu_{\alpha,\mathbb{Q}}(x)$ *must* be irreducible.

(c) If $A \in \operatorname{Mat}_n(\mathbb{R})$, then its minimal polynomial $\mu_A(x)$ *must* be irreducible.

Solution:

(a) To start, we have that $\operatorname{Gal}(E/\mathbb{Q})$ acts transitively on the roots of $f(x)$; since there are at most n roots for $f(x)$ in its splitting field, it follows that $\operatorname{Gal}(E/\mathbb{Q}) \hookrightarrow S_m$, where m is the number of distinct roots of $f(x)$ in E. Yet, since $\operatorname{Gal}(E/\mathbb{Q}) = S_n$, it follows that $f(x)$ has n distinct roots in E, whence it is separable.

Since $f(x)$ is a separable polynomial whose Galois group is S_n, its Galois group is a transitive subgroup of S_n, whence $f(x)$ is irreducible.

Problem (Algebra II HW 11, Problem 6): Let p be a prime, $K = \mathbb{F}_p(t)$, the field of rational functions in one variable over $\mathbb{F}_p$. Let $\sigma\colon K \to K$ be the unique automorphism of K such that $\sigma(t) = t+1$, and $G = \langle\sigma\rangle$. Observe that G is cyclic of order p. Let $F = K^G$. Find an explicit element s such that $F = \mathbb{F}_p(s)$, and prove your answer.

Solution: Consider the product given by

$$s = \prod_{i=0}^{p-1}(t+i),$$

so that

$$\begin{aligned}\sigma(s) &= \prod_{i=0}^{p-1}(\sigma(t)+i)\\ &= \prod_{i=0}^{p-1}(t+i+1)\\ &= s\end{aligned}$$

since σ has order p. Therefore, $s \in K^G$, so that $\mathbb{F}_p(s) \subseteq K^G$.

Consider now the tower of extensions given by $\mathbb{F}_p(s) \subseteq K^G \subseteq K$. We may write

$$K = \left(\mathbb{F}_p(s)\right)(t),$$

so we may compute

$$\begin{aligned}\left[K : \mathbb{F}_p(s)\right] &= \left[\mathbb{F}_p(s)(t) : \mathbb{F}_p(s)\right]\\ &\leq p,\end{aligned}$$

where the bound appears from the fact that t is a root of the polynomial

$$\prod_{i=0}^{p-1}(x+i) - s \in \mathbb{F}_p(s)[x].$$

Meanwhile, by Artin's Lemma, we know that K/K^G is Galois with Galois group G, whence $\left[K : K^G\right] = |G| = p$. Hence, we have $\left[K^G : \mathbb{F}_p(s)\right] = 1$, so that $\mathbb{F}_p(s) = K^G$.

Problem (Algebra II HW 11, Problem 1)**:** Let p be a prime, n a positive integer, and let $\Phi_n(x) = x^{p^n} - x$ be an element of $\mathbb{F}_p[x]$. Prove that Φ_n is equal to the product of all the monic irreducible polynomials in $\mathbb{F}_p[x]$ whose degrees divide n, with each polynomial appearing with multiplicity one.

Solution: First, we observe that if $\overline{\mathbb{F}_p}$ is an algebraic closure, then the splitting field of $\Phi_n(x)$ is given by $\mathbb{F}_{p^n} \subseteq \overline{\mathbb{F}_p}$. We may write

$$\Phi_n(x) = \prod_{\alpha\in\mathbb{F}_{p^n}}(x-\alpha)$$

split in $\overline{\mathbb{F}_p}$. Then, any irreducible factor f_i of Φ_n will define a subfield of the splitting field of $\Phi_n(x)$, which is given by $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$ for some n. We have that $m|n$, so that $\deg(f_i)|n$ and that

$$f_i(x) = \prod_{\alpha\in\mathbb{F}_{p^m}}(x-\alpha)$$

fully splits in $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$. In particular, this means that $f_i(x)$ has multiplicity one in the product decomposition of $\Phi_n(x)$.

Similarly, if $f_m(x)$ is a monic irreducible polynomial in $\mathbb{F}_p[x]$ with degree $m|n$, then $f_m(x)$ defines a field extension $\mathbb{F}_p \subseteq F \subseteq \mathbb{F}_{p^n} \subseteq \overline{\mathbb{F}_p}$ with $\left[F : \mathbb{F}_p\right] = m$, whence $F = \mathbb{F}_{p^m}$. Since $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$, it follows that every root of f_m in $\overline{\mathbb{F}_p}$ is a root in $\Phi_n(x)$, whence $f_m(x)|\Phi_n(x)$, and $f_m(x)$ has multiplicity one by the earlier discussion. Therefore,

$$\Phi_n(x) = \prod_{m|n} f_m(x),$$

where $f_m(x)$ is a monic irreducible factor of degree m for $\Phi_n(x)$ and each f_m appears with multiplicity one.

Problem (January '20, Problem 8)**:** Let K/F be a field extension. Suppose that $K = F(a,b)$ for some $a, b \in K$ such that $a^2 \in F$ and $b^2 \in F$.

(a) Prove that $[K : F] \leq 4$.

(b) Give an example (with full proof) where $[K : F] = 4$.

(c) Assume that $\operatorname{char}(F) \neq 2$. Prove that the extension K/F is Galois.

(d) Now, assume that F is finite. Prove that $[K : F] \leq 2$.

Solution:

(a) We have

$$[F(a,b) : F] = [F(a,b) : F(b)][F(b) : F].$$

Observe that $\mu_{a,F(b)}|\mu_{a,F}$, with $\mu_{a,F}$ and $\mu_{b,F}$ having degree 2. Therefore, we have

$$\begin{aligned}[F(a,b) : F] &= [F(a,b) : F(b)][F(b) : F]\\ &\leq 4.\end{aligned}$$

(b) We consider the case of $\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right)$. We start by noting that 2 and 3 are coprime. Observe that if $a + b\sqrt{2} = \sqrt{3}$ with $a, b \in \mathbb{Q}$, then we get

$$3 = a^2 + 2b^2 + 2ab\sqrt{2},$$

giving that $\sqrt{2}$ is rational, which is a contradiction. Therefore, we have that $\sqrt{3}$ is not contained in $\mathbb{Q}\left(\sqrt{2}\right)$, whence the minimal polynomial is still degree 2. Therefore, we get

$$\begin{aligned}\left[\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right) : \mathbb{Q}\right] &= \left[\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right) : \mathbb{Q}\left(\sqrt{2}\right)\right]\left[\mathbb{Q}\left(\sqrt{2}\right) : \mathbb{Q}\right]\\ &= 4.\end{aligned}$$

(c) We observe that a and b are roots of the separable polynomials $x^2 - a^2$ and $x^2 - b^2$ respectively. We have that $F(a,b) = F(a)F(b)$ can be expressed as a compositum, and that $F(a)/F$ and $F(b)/F$ are separable extensions of degree 2, hence they are Galois extensions. Since the compositum of two Galois extensions is Galois, we get that K/F is Galois.

(d) If F is finite, then $F = \mathbb{F}_q$ for some $q = p^m$. Since $F(a)/F$ and $F(b)/F$ are field extensions of degree at most 2, the result follows if either $F(a) = F$ or $F(b) = F$. Else, if both $F(a)/F$ and $F(b)/F$ are extensions of degree 2, then by uniqueness of finite fields, they are both equal to the field $\mathbb{F}_{q^2}$, whence $[F(a,b) : F] \leq 2$.

Galois Theory Notes

General Galois Theory Notes

Theorem: If $f(x) \in K[x]$ is a separable polynomial of degree n, then f is irreducible if and only if the Galois group $\operatorname{Gal}(\operatorname{spl}_K(f(x))/K)$ is a transitive subgroup of S_n.

Proof. If $f(x)$ is irreducible, then we know that the Galois group acts transitively on the roots.

Meanwhile, if $f(x)$ is reducible, then $f(x)$ is a product of distinct irreducibles since it is separable. Let r_i and r_j be roots of different irreducible factors of $f(x)$. Then, for each $\sigma \in \operatorname{Gal}(\operatorname{spl}_K(f(x))/K)$, we have $\sigma(r_i)$ has the same minimal polynomial over K as r_i, so we cannot have $\sigma(r_i) = r_j$, whence the Galois group is not transitive. □

Galois Groups of Polynomials

Definition: Let $x_1, \dots, x_n$ be indeterminates. The *elementary symmetric functions* $s_1, \dots, s_n$ are defined by

$$\begin{aligned} s_1 &= x_1 + x_2 + \cdots + x_n \\ s_2 &= \prod_{i<j} x_i x_j \\ &\vdots \\ s_n &= x_1 x_2 \cdots x_n. \end{aligned}$$

The *general polynomial* of degree n is the polynomial

$$g(x) = (x - x_1)(x - x - 2) \cdots (x - x_n)$$

with roots $x_1, \dots, x_n$.

Observe that the general polynomial can be written as

$$(x - x_1)(x - x_2) \cdots (x - x_n) = x^n - s_1 x^{n-1} + \cdots + (-1)^n s_n.$$

This means that if F is some field, then the extension $F(x_1, \dots, x_n)$ is a Galois extension of the field $F(s_1, \dots, s_n)$ since it is the splitting field of the general polynomial of degree n. Furthermore, $F(s_1, \dots, s_n)$ is contained in the fixed field of the action of S_n on $F(x_1, \dots, x_n)$, where $\sigma \cdot x_i = x_{\sigma(i)}$ for any $\sigma \in S_n$. Since the fixed field of S_n has index $n!$ in $F(x_1, \dots, x_n)$ by the fundamental theorem of Galois theory, it follows that we have

$$[F(x_1, \dots, x_n) : F(s_1, \dots, s_n)] \leq n!,$$

whence $F(s_1, \dots, s_n)$ is the full fixed field of $F(x_1, \dots, x_n)$.

Theorem: The general polynomial

$$x^n - s_1 x^{n-1} + s_2 x^{n-2} + \cdots + (-1)^n s_n$$

over the field $F(s_1, \dots, s_n)$ is separable with Galois group S_n.

Now, we will discuss the discriminant, which is an essential tool in understanding the Galois groups of polynomials, especially those of degrees two, three, and four.

Definition: The *discriminant* of $x_1, \dots, x_n$ is given by

$$\Delta = \prod_{i<j} \left(x_i - x_j\right)^2.$$

The discriminant of a polynomial is the discriminant of the roots of the polynomial.

We start by observing that the discriminant is a symmetric function in $x_1, \dots, x_n$, so it is an element of the field $K = F(s_1, \dots, s_n)$.

Recalling the definition of the alternating group A_n, we have that $\sigma \in S_n$ is in A_n if and only if σ fixes

$$\sqrt{\Delta} = \prod_{i<j} \left(x_i - x_j\right)$$

in $\mathbb{Z}[x_1, \dots, x_n]$. Notice then that if F has characteristic not equal to 2, then $\sqrt{\Delta}$ generates the fixed field of A_n, which is a quadratic extension of K.

If $f(x)$ is a monic degree n polynomial with roots $\alpha_1, \dots, \alpha_n$, then the discriminant is given by

$$\Delta = \prod_{i<j} \left(\alpha_i - \alpha_j\right)^2.$$

Observe that $\Delta \in F$ since Δ is fixed by the Galois group. Furthermore, $\Delta = 0$ if and only if $f(x)$ is separable, so if F is perfect (i.e., characteristic zero or finite), then $f(x)$ is reducible if and only if $\Delta = 0$ since every irreducible polynomial over a perfect field is separable. Additionally, observe that $\sqrt{\Delta} \in F$ if and only if the Galois group of $f(x)$ is contained in A_n.

Now, we start our investigation with the polynomials of degree 2. The discriminant of a polynomial $x^2 + ax + b$ is given by $\Delta = (\alpha - \beta)^2$, which can be written as

$$\begin{aligned} \Delta &= s_1^2 - 4s_2 \\ &= a^2 - 4b. \end{aligned}$$

The polynomial is separable if and only if $a^2 - 4b \neq 0$, and the Galois group is a subgroup of $S_2 = \mathbb{Z}/2\mathbb{Z}$. This means that the Galois group is trivial if and only if $a^2 - 4b$ is a rational square.

Next, we consider the cubic polynomials. If we have

$$f(x) = x^3 + ax^2 + bx + c,$$

which upon the substitution $x = y - a/3$, yields a reduced cubic

$$g(y) = y^3 + py + q,$$

where

$$\begin{aligned} p &= \frac{1}{3}\left(3b - a^2\right) \\ q &= \frac{1}{27}\left(2a^3 - 9ab + 27c\right). \end{aligned}$$

Since the roots of the polynomials f and g differ by the constant $a/3$, and the formula for the discriminant involves the difference of these roots, it follows that the polynomials have the same discriminant.

Letting α, β, γ be the roots of the polynomial $g(y)$, then we have

$$\begin{aligned} g'(y) &= (y - \alpha)(y - \beta) + (y - \alpha)(y - \gamma) + (y - \beta)(y - \gamma) \\ g'(\alpha) &= (\alpha - \beta)(\alpha - \gamma) \\ g'(\beta) &= (\beta - \alpha)(\beta - \gamma) \\ g'(\gamma) &= (\gamma - \alpha)(\gamma - \beta). \end{aligned}$$

Therefore, we have

$$\begin{aligned} \Delta &= ((\alpha - \beta)(\alpha - \gamma)(\beta - \gamma))^2 \\ &= -g'(\alpha)g'(\beta)g'(\gamma). \end{aligned}$$

Since $g'(y) = 3y^2 + p$, it follows that

$$\begin{aligned}-\Delta &= \left(3\alpha^2 + p\right)\left(3\beta^2 + p\right)\left(3\gamma^2 + p\right)\\ &= 27\alpha^2\beta^2\gamma^2 + 9p\left(\alpha^2\beta^2 + \alpha^2\gamma^2 + \beta^2\gamma^2\right) + 3p^2\left(\alpha^2 + \beta^2 + \gamma^2\right) + p^3.\end{aligned}$$

These expressions are elementary symmetric functions of the roots, where for g, $s_1 = 0$, $s_2 = p$, and $s_3 = -q$, whence

$$\Delta = -4p^3 - 27q^2.$$

Therefore, expressing Δ in terms of a, b, c, we have

$$\Delta = a^2b^2 - 4b^3 - 4a^3c - 27c^2 + 18abc.$$

This allows us to compute the Galois group as follows.

(i) If $f(x)$ is reducible, then either $f(x)$ splits into three linear factors or into a linear factor and an irreducible quadratic. In the first case, the Galois group is trivial and in the second case, the Galois group is of order 2.

(ii) If $f(x)$ is irreducible, then a root of $f(x)$ generates a degree 3 extension over F. Therefore, the degree of the splitting field over F is divisible by 3.

Since the Galois group is a subgroup of S_3, it follows that there are only two options, either A_3 or S_3. In particular, the Galois group is A_3 if and only if the discriminant Δ is a square.

If Δ is a square, then the splitting field of the irreducible cubic $f(x)$ is obtained by adjoining any root of $f(x)$ to F. If Δ is not the square of an element of F, then the splitting field of $f(x)$ is of degree 6 over F, given by $F\left(\theta, \sqrt{\Delta}\right)$ where θ is some root of $f(x)$. This extension is Galois over F generated by the automorphisms σ and τ, where σ takes θ to any other root of $f(x)$ fixing $\sqrt{\Delta}$ and τ takes $\sqrt{\Delta} \mapsto -\sqrt{\Delta}$ and fixing θ.

Finally, we discuss the Galois group of a quartic polynomial

$$f(x) = x^4 + ax^3 + bx^2 + cx + d.$$

Then, by substituting with $x = y - a/4$, this yields the quartic

$$g(y) = y^4 + py^2 + qy + r$$

with

$$\begin{aligned}p &= \frac{1}{8}\left(-3a^2 + 8b\right)\\ q &= \frac{1}{8}\left(a^3 - 4ab - 8c\right)\\ r &= \frac{1}{256}\left(-3a^4 + 16a^2b - 64ac + 256d\right).\end{aligned}$$

Let $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ be the roots of $g(y)$ and G the Galois group for the splitting field of $g(y)$.

If $g(y)$ is reducible, then the following cases hold.

(i) If $g(y)$ splits into a linear and a cubic, then G is the Galois group of the cubic.

(ii) If $g(y)$ splits into two irreducible quadratics, then the splitting field is $F\left(\sqrt{\Delta_1}, \sqrt{\Delta_2}\right)$, where Δ_1 and Δ_2 are the discriminants of the quadratics.

If Δ_1 and Δ_2 do not differ by a square, then G is isomorphic to a Klein 4-subgroup of S_4.

If Δ_1 is a square multiplied by Δ_2, then $F\left(\sqrt{\Delta_1}, \sqrt{\Delta_2}\right) = F\left(\sqrt{\Delta_1}\right)$ is a quadratic extension with G isomorphic to $\mathbb{Z}/2\mathbb{Z}$.

Now, we have the case of $g(y)$ irreducible.Then, the Galois group is transitive on its roots. There are five transitive subgroups (up to conjugation) of S_4, given by copies of S_4, A_4, D_4, V, C, where V denotes the Klein 4-group and C denotes the copy of $\mathbb{Z}/4\mathbb{Z}$.

We consider the elements

$$\begin{aligned}\theta_1 &= (\alpha_1+\alpha_2)(\alpha_3+\alpha_4)\\ \theta_2 &= (\alpha_1+\alpha_3)(\alpha_2+\alpha_4)\\ \theta_3 &= (\alpha_1+\alpha_4)(\alpha_2+\alpha_3)\end{aligned}$$

inside the splitting field for $g(y)$. Notice that the elements are permuted among themselves by the permutations of S_4. The stabilizer of θ_1 is the dihedral group D_4, while the stabilizer of all three of θ_1, θ_2, and θ_3 is the copy of V.

The elementary symmetric functions in the θ_i are fixed by S_4, so they are in F. These elementary symmetric functions evaluated on $\theta_1,\dots,\theta_3$ are given by $2p$, p^2-4r, and $-q^2$. This gives that θ_1, θ_2, and θ_3 are the roots of

$$h(x) = x^3 - 2px^2 + \left(p^2-4r\right)x + q^2.$$

This is known as the *resolvent cubic* for the quartic $g(y)$. It can be shown (see DF p.614) that the discriminant Δ of the resolvent cubic is equal to the resolvent of $g(y)$. In particular, this gives

$$\Delta = 16p^4r - 4p^3q^2 - 128p^2r^2 + 144pq^2r - 27q^4 + 256r^3.$$

Remark: In the special case where $g(y) = y^4 + py^2 + r$ (which emerges during analysis of quartics whose roots are nested square roots), we have

$$\Delta = 16p^4r - 128p^2r^2 + 256r^3.$$

In particular, since the splitting field for the resolvent cubic is a subfield of the splitting field of the original quartic, the Galois group for $h(x)$ is a quotient of G, so knowing the action of the Galois group of $h(x)$ on the roots of $h(x)$ gives information about the Galois group of $g(y)$.

(i) If the resolvent cubic is irreducible, and Δ is not a square, then G is not contained in A_4 with the Galois group of the resolvent cubic equal to S_3. In particular, this means that the degree of the splitting field for $g(y)$ is divisible by 6, so that $G = S_4$.

(ii) If the resolvent cubic is irreducible, and Δ is a square, then G is a subgroup of A_4 and 3 divides the order of G since the Galois group of the resolvent cubic is A_3, so that $G = A_4$.

(iii) If the resolvent cubic is reducible, and $h(x)$ splits completely, then every element of G fixes the elements θ_1, θ_2, and θ_3, so $G = V$.

(iv) If the resolvent cubic splits into a linear factor and a quadratic, then G stabilizes θ_1 but not θ_2 or θ_3, so $G \subseteq D_4$ and $G \not\subseteq V$, so $G = D_4$ or $G = C$.

Now, notice that $F\left(\sqrt{\Delta}\right)$ is the fixed field of the elements of $G \cap A_4$. We have $D_4 \cap A_4 = V$ and $C \cap A_4 = \{e, (1,3)(2,4)\}$, the latter of which is not transitive. Therefore, we have the case $D_4 \cap A_4 = V$ if and only if $g(y)$ is irreducible over $F\left(\sqrt{\Delta}\right)$.

Solvability by Radicals

Definition: An extension K/F is called *cyclic* if it is Galois with cyclic Galois group.

Proposition: Suppose F is a field of characteristic not dividing n that contains the nth roots of unity. Then, the extension $F\left(\sqrt[n]{a}\right)$, for some $a \in F$, is cyclic over F with degree dividing n.

Proof. The extension $K = F\left(\sqrt[n]{a}\right)$ is Galois over F if F contains the nth roots of unity, since it is the splitting field for the polynomial $x^n - a$. For any $\sigma \in \mathrm{Gal}(K/F)$, we have $\sigma\left(\sqrt[n]{a}\right)$ is another root of this polynomial, so $\sigma\left(\sqrt[n]{a}\right) = \zeta_\sigma \sqrt[n]{a}$ for some nth root of unity ζ_σ.

We thus get a map $\mathrm{Gal}(K/F) \to \mu_n$ taking $\sigma \mapsto \zeta_\sigma$, where μ_n denotes the group of nth roots of unity. Since F contains μ_n, it follows that every nth root of unity is fixed by every element of $\mathrm{Gal}(K/F)$, so that $\zeta_{\sigma\tau} = \zeta_\sigma\zeta_\tau$. In particular, this means that the map is a homomorphism, so there is an injection of $\mathrm{Gal}(K/F)$ into the cyclic group μ_n with order n. □

Now, assume that K is a cyclic extension of degree n over a field F with characteristic not dividing n that contains the nth roots of unity. Let σ be a generator for the cyclic group $\mathrm{Gal}(K/F)$.

Definition: Let $\alpha \in K$ and ζ an nth root of unity. Define the *Lagrange resolvent* (α, ζ) by

$$(\alpha, \zeta) = \alpha + \zeta\sigma(\alpha) + \zeta^2\sigma^2(\alpha) + \cdots + \zeta^{n-1}\sigma^{n-1}(\alpha).$$

Note that by applying σ to (α, ζ), we get

$$\sigma(\alpha, \zeta) = \zeta^{-1}(\alpha, \zeta).$$

In particular, this means that $(\alpha, \zeta)^n$ is fixed by $\mathrm{Gal}(K/F)$, so it is an element of F for any $\alpha \in K$.

If ζ is a primitive nth root of unity, then since the automorphisms $\mathrm{id}, \sigma, \dots, \sigma^{n-1}$ are linearly independent, there is an element $\alpha \in K$ with $(\alpha, \zeta) \neq 0$. Therefore, we have

$$\sigma^i(\alpha, \zeta) = \zeta^{-i}(\alpha, \zeta),$$

so it follows that σ^i does not fix (α, ζ) for $1 \leq i < n$. Therefore, the element cannot lie in any proper subfield of K, whence $K = F((\alpha, \zeta))$. Since $(\alpha, \zeta)^n = a \in F$, it follows that $F\left(\sqrt[n]{a}\right) = F((\alpha, \zeta)) = K$. Therefore, we have the following.

Proposition: Any cyclic extension of degree n over a field F of characteristic not dividing n which contains the nth roots of unity is of the form $F\left(\sqrt[n]{a}\right)$ for some $a \in F$.

A group G is said to have *exponent n* if $g^n = 1$ for each $g \in G$. If F is a field of characteristic not dividing n that contains the nth roots of unity, then if $a_1, \dots, a_k \in F^\times$, we observe that the extension

$$K = F(\sqrt[n]{a_1}, \dots, \sqrt[n]{a_k})$$

is an abelian extension of F whose Galois group is of exponent n. Furthermore, any abelian extension of exponent n is of this form. The Galois group is isomorphic to the group generated by the elements $a_1, \dots, a_k$ in the group $F^\times/(F^\times)^n$, and any two such extensions are equal if and only if their associated groups are equal.

From now on, we will consider base fields of characteristic 0. The results are valid over fields whose characteristics do not divide the orders of the roots taken.

Definition: An element α that is algebraic over F can be *solved in terms of radicals* if α is an element of a field K that can be obtained by a succession of simple radical extensions

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_{i+1} \subseteq K_i \subseteq K_s = K,$$

where $K_{i+1} = K_i\left(\sqrt[n_i]{a_i}\right)$ for some $a_i \in K_i$, where $\sqrt[n_i]{a_i}$ denotes some root of the polynomial $x^{n_i} - a_i$. Such a field K is called a *root extension* of F.

A polynomial $f(x) \in F[x]$ is said to be solved by radicals if all its roots can be solved for in terms of radicals.

Lemma: Suppose α is contained in a root extension K of F. Then, α is contained in a root extension which is Galois over F where each extension K_{i+1}/K_i is cyclic.

Proof. Let L be the Galois closure of K over F. For any $\sigma \in \mathrm{Gal}(L/F)$, we have the chain of subfields

$$F = \sigma K_0 \subseteq \sigma K_1 \subseteq \cdots \subseteq \sigma K_i \subseteq \sigma K_{i+1} \subseteq \cdots \subseteq \sigma K_s = \sigma K,$$

where $\sigma K_{i+1}/\sigma K_i$ is a simple radical extension since it is generated by the element $\sigma\left(\sqrt[n_i]{a_i}\right)$, which is a root of the equation $x^{n_i} - \sigma(a_i)$ over $\sigma(K_i)$.

The composite of two root extensions is again a root extension, since if K' is a root extension with subfields K_i', we may take the composite of K_1' with the fields $K_0, K_1, \dots, K_s$, then the composite with K_2', etc., so that each extension is a simple radical extension.

It follows that the composite of all the conjugate fields $\sigma(K)$ for all $\sigma \in \mathrm{Gal}(L/F)$ is again a root extension. Yet, since this field is L, it follows that α is contained in a Galois root extension.

Now, adjoin to F the n_ith roots of unity for all the roots $\sqrt[n_i]{a_i}$ of the simple radical extensions of the Galois root extension K/F. This gives the field F'. Form the composite of F' with the root extension

$$F \subseteq F' = F'K_0 \subseteq F'K_1 \subseteq \cdots \subseteq F'K_i \subseteq F'K_{i+1} \subseteq \cdots \subseteq F'K_s = F'K.$$

The field $F'K$ is a Galois extension of F since it is the composite of two Galois extensions. The extension from F to $F' = F'K_0$ can be given as a chain of subfields with cyclic extensions (since this holds for any abelian extension). Each extension $F'K_{i+1}/F'K_i$ is a simple radical extension, and since there are roots of unity in the base fields, it follows that each individual extension from F' to $F'K$ is a cyclic extension, so $F'K/F$ is a root extension which is Galois over F with cyclic intermediate extensions. $\square$

Recall that a finite group G is called *solvable* if there is a subgroup series

$$1 = G_s \trianglelefteq G_{s-1} \trianglelefteq \cdots \trianglelefteq G_{i+1} \trianglelefteq G_i \trianglelefteq G_0 = G$$

where G_i/G_{i+1} is cyclic.[III] Note that subgroups and quotients of solvable groups are solvable, and that if $H \trianglelefteq G$ and G/H are both solvable, then G is solvable.

Theorem: The polynomial $f(x)$ can be solved by radicals if and only if its Galois group is a solvable group.

Proof. Suppose $f(x)$ can be solved by radicals. Then, each root of $f(x)$ is contained in a root extension, and the composite L of such extensions is another root extension. Let G_i be the subgroups of the Galois group corresponding to the subfields K_i where $i = 0, 1, \dots, s-1$. Since

$$\mathrm{Gal}(K_{i+1}/K_i) = G_i/G_{i+1}$$

by the Galois correspondence. Each such group is cyclic since K_{i+1}/K_i is a root extension. It follows that the Galois group $G = \mathrm{Gal}(L/F)$ is a solvable group, and since the Galois group of $f(x)$ is a quotient of the solvable group G, it follows that the Galois group is solvable.

Now, suppose that the Galois group G of $f(x)$ is solvable, and let K be the splitting field of $f(x)$. Taking fixed fields of the subgroups in the series, we have a chain

$$F = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_i \subseteq K_{i+1} \subseteq \cdots \subseteq K_s = K,$$

where K_{i+1}/K_i is a cyclic extension of degree n_i. If F' is the cyclotomic field of all roots of unity with order n_i, we get a sequence of extensions $K_i' = F'K_i$ given by

$$F \subseteq F' = F'K_0 \subseteq K_1' \subseteq \cdots \subseteq K_i' \subseteq K_{i+1}' \subseteq \cdots \subseteq K_s' = F'K.$$

The extension $F'K_{i+1}/F'K_i$ is cyclic of degree n_i. Since there are the appropriate roots of unity in the base fields, it follows that each of these cyclic extensions is a simple radical extension, so each of the roots of $f(x)$ is contained in the root extension $F'K$, so that $f(x)$ is solvable by radicals. $\square$

Theorem: The general quintic is not solvable by radicals.

Proof. The general quintic has Galois group S_5, and S_5 is not solvable. $\square$

We will now discuss the cases where a quintic *is* solvable. This is equivalent to finding solvable subgroups of S_5.

[III] Equivalently, the derived series $D^{(i)}(G)$ terminates at the identity. See the group theory notes for an explanation of the equivalence.

Galois Theory Worked Solutions

Problem (August '89, Problem 8): Let $\beta = \sqrt{2+\sqrt{2}}$. Show $\mathbb{Q}(\beta)/\mathbb{Q}$ is Galois with G cyclic; set up the Galois correspondence.

Solution: Working over the algebraic numbers, we observe that $\sqrt{2+\sqrt{2}}$ is a root of the irreducible polynomial

$$f(x) = x^4 - 4x^2 + 2.$$

The discriminant is given by

$$\begin{aligned}\Delta &= 16(-4)^4(2) - 128(-4)^2(2)^2 + 256(2)^3\\ &= 2^{11},\end{aligned}$$

so that $\mathbb{Q}\left(\sqrt{\Delta}\right) = \mathbb{Q}\left(\sqrt{2}\right)$. Then, over $\mathbb{Q}\left(\sqrt{2}\right)$, we have

$$f(x) = \left(\left(x^2-2\right)-\sqrt{2}\right)\left(\left(x^2-2\right)+\sqrt{2}\right),$$

which means that $f(x)$ is not irreducible over $\mathbb{Q}\left(\sqrt{2}\right)$. In particular, this means that the Galois group is equal to $\mathbb{Z}/4\mathbb{Z}$, and in particular, since the degree of $\mathbb{Q}(\beta)$ over $\mathbb{Q}$ is 4, it follows that the extension is indeed Galois.

Problem (January '17, Problem 1): Consider the polynomial $f(x) = x^4 - 2x^2 - 6$. Prove this polynomial is irreducible. Describe the splitting field of this polynomial over $\mathbb{Q}$, and the Galois group of this splitting field.

Solution: We observe that this polynomial is contained in $\mathbb{Z}[x]$, so viewing $f(x)$ as an element of $\mathbb{Z}[x]$, we have that $f(x)$ is monic and $6 \notin \left(2^2\right)$, where $(2) \subseteq \mathbb{Z}$ is the ideal generated by the prime 2. By Eisenstein's criterion, it follows that $f(x)$ is irreducible over $\mathbb{Z}$, and by Gauss's Lemma, $f(x)$ is irreducible over $\mathbb{Q}$. Let K denote the splitting field of $f(x)$ over $\mathbb{Q}$.

Considering the set of roots

$$W = \left\{\sqrt{1+\sqrt{7}}, -\sqrt{1+\sqrt{7}}, \sqrt{1-\sqrt{7}}, -\sqrt{1-\sqrt{7}}\right\},$$

we observe that if we let

$$L = \mathbb{Q}\left(\sqrt{1+\sqrt{7}}\right),$$

then $f(x)$ is reducible over L, but has only two roots in L, so that $[K : L] = 2$. Since $[L : \mathbb{Q}] = 4$, it follows that $[K : \mathbb{Q}] = 8$.

Since $f(x)$ is irreducible, it follows that the Galois group of $f(x)$ acts transitively on the roots of f, and has order 8, so that $G \cong D_8$.

Problem (August '02, Problem 9): Construct a Galois extension of $\mathbb{Q}$ of degree 3.

Solution: Consider ζ_7, a primitive 7th root of unity over $\mathbb{Q}$. Then, we have that $\mathbb{Q}(\zeta_7)/\mathbb{Q}$ is a Galois extension, since $\mathbb{Q}(\zeta_7)$ is the splitting field of the separable polynomial $x^7 - 1$, and has $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q}) \cong (\mathbb{Z}/7\mathbb{Z})^\times \cong \mathbb{Z}/6\mathbb{Z}$. A Galois extension of degree 3 corresponds to a subgroup of index 3 in $(\mathbb{Z}/7\mathbb{Z})^\times$, which is the unique subgroup of order 2 in $(\mathbb{Z}/7\mathbb{Z})^\times$. If we let $\langle\sigma\rangle \subseteq \mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$ be this subgroup, then σ is given by an element of order 2 in $\mathrm{Gal}(\mathbb{Q}(\zeta_7)/\mathbb{Q})$; since complex conjugation is a $\mathbb{Q}$-automorphism of $\mathbb{Q}(\zeta_7)$, it follows that σ is complex conjugation.

Observe now that

$$\sum_{i=0}^{1} \sigma^i(\zeta_7) = \zeta_7 + \zeta_7^{-1}$$

is invariant under σ, so that $\mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right) \subseteq \mathbb{Q}(\zeta_7)^{\sigma}$. Furthermore, letting $\alpha = \zeta_7 + \zeta_7^{-1}$, we have

$$\zeta_7^2 - \alpha\zeta_7 + 1 = 0,$$

whence $\mu_{\zeta_7,\mathbb{Q}(\alpha)}(x)$ has degree at most 2.

Therefore, we observe that $\mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right)$ is an extension with $\left[\mathbb{Q}(\zeta_7) : \mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right)\right] = 2$, whence $\mathbb{Q}\left(\zeta_7 + \zeta_7^{-1}\right)$ is the desired extension of degree 3 over $\mathbb{Q}$.

Problem (August '08, Problem 8):

(a) Find a splitting field E of the polynomial $x^4 + 3x^3 + 4x^2 + 3x + 3$ over $F = \mathbb{Q}$, and find its degree $[E : F]$.

(b) Find the Galois group $\operatorname{Gal}(E/F)$ of the extension field.

(c) Diagram the lattice of subgroups of the Galois group and the corresponding lattice of intermediate subfields of E/F.

Solution:

(a) Taking the factorization over $\mathbb{Q}$, we have that the polynomial factorizes as $\left(x^2+1\right)\left(x^2+3x+3\right)$. Over $\mathbb{C}$, we may factorize this as

$$p(x) = (x-i)(x+i)\left(x - \left(-\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)\right)\left(x - \left(-\frac{3}{2} - i\frac{\sqrt{3}}{2}\right)\right).$$

Adjoining the roots to $\mathbb{Q}$ gives that $E \subseteq \mathbb{Q}\left(i, \sqrt{3}\right) \subseteq E$. Therefore, $E = \mathbb{Q}\left(i, \sqrt{3}\right)$. We have that

$$\begin{aligned}[E : \mathbb{Q}] &= \left[\mathbb{Q}\left(i, \sqrt{3}\right) : \mathbb{Q}(i)\right][\mathbb{Q}(i) : \mathbb{Q}] \\ &= 4,\end{aligned}$$

where we use the fact that $\mu_{\sqrt{3},\mathbb{Q}(i)} = x^2 - 3$ is irreducible over $\mathbb{Q}(i)$ as 3 is not a square mod 4.

(b) We observe that the maps

$$\sigma = \begin{cases} i \mapsto -i \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$$

$$\tau = \begin{cases} i \mapsto i \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$$

are automorphisms of $\mathbb{Q}\left(i, \sqrt{3}\right)$ that determine the order 4 group $\operatorname{Gal}(E/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

(c) The fixed field for the group $\langle\sigma\rangle$ is $\mathbb{Q}\left(\sqrt{3}\right)$, while the fixed field for the group $\langle\tau\rangle$ is $\mathbb{Q}(i)$.

Problem (August '18, Problem 7): What is the Galois group of $x^5 - 1$ over a field with 7 elements?

Solution: We seek to understand the factorization into irreducibles of $x^5 - 1$. Observe that we have

$$x^5 - 1 = (x-1)\left(x^4 + x^3 + x^2 + x + 1\right),$$

where we observe that the latter factor is equal to $\Phi_5(x)$, the 5th cyclotomic polynomial. Recalling the theorem that the nth cyclotomic polynomials factor over $\mathbb{F}_p$ into irreducibles whose degree is equal to the order of p modulo n, we observe that $7 \equiv 2 \bmod 5$, and 2 has order 4 in $(\mathbb{Z}/5\mathbb{Z})^\times$, so that $\Phi_5(x)$ is irreducible over $\mathbb{F}_7$.

In particular, this means that the Galois group of $x^5 - 1$ is equal to the Galois group of the splitting field; since $[\mathbb{F}_7(\alpha) : \mathbb{F}_7] = 4$, it follows that the Galois group is isomorphic to $\mathbb{Z}/4\mathbb{Z}$.

Problem (August '20, Problem 8)**:** Let K be the splitting field of the polynomial $f(x) = \left(x^3 - 11\right)\left(x^2 + x - 1\right)$ over $\mathbb{Q}$.

(a) Find (with proof) the degree of the extension $K/\mathbb{Q}$.

(b) Determine the isomorphism class of the Galois group $\operatorname{Gal}(K/\mathbb{Q})$, and prove your answer.

Solution:

(a) Considering the embedding of $\mathbb{Q}[x] \subseteq \mathbb{C}[x]$, we observe that the roots of $f(x)$ are given by

$$A = \left\{\sqrt[3]{11}, \omega\sqrt[3]{11}, \omega^2\sqrt[3]{11}, \frac{-1+\sqrt{5}}{2}, \frac{-1-\sqrt{5}}{2}\right\},$$

where ω is a primitive cube root of unity. In particular, this means that the splitting field K of $f(x)$ is contained in $F = \mathbb{Q}\left(\omega, \sqrt[3]{11}, \sqrt{5}\right)$. Meanwhile, we have $F \subseteq K$ since we may find $\omega = \frac{\omega\sqrt[3]{11}}{\sqrt[3]{11}}$ and $\sqrt{5} = 1 + 2\frac{-1+\sqrt{5}}{2}$, so that $K = \mathbb{Q}\left(\omega, \sqrt[3]{11}, \sqrt{5}\right)$.

To compute the degree, we observe that the minimal polynomials are given by

$$\begin{aligned} \mu_{\sqrt[3]{11},\mathbb{Q}}(x) &= x^3 - 11 \\ \mu_{\omega,\mathbb{Q}}(x) &= x^2 + x + 1 \\ \mu_{\sqrt{5},\mathbb{Q}} &= x^2 - 5. \end{aligned}$$

Computing the degree, we observe that $\mu_{\sqrt[3]{11},\mathbb{Q}(\omega,\sqrt{5})} = \mu_{\sqrt[3]{11}}$, and similarly for $\mu_{\omega,\mathbb{Q}(\sqrt{5})}$ with $\mu_{\omega,\mathbb{Q}}$, so that the degree of $K/\mathbb{Q}$ is 12.

(b) First, we recall the theorem that a separable polynomial $f(x)$ is irreducible if and only if the action of the Galois group on the roots of $f(x)$ is transitive. Since f is separable (as we are working over $\mathbb{Q}$) and not irreducible, it follows that the Galois group of $f(x)$ is not transitive.

Now, we observe that $f(x)$ has two irreducible factors — namely, $p(x) = x^3 - 11$ (which is irreducible by the Eisenstein criterion) and $q(x) = x^2 + x - 1$, and in particular, the extension $K/\mathbb{Q}$ is equal to the compositum of the splitting fields of $p(x)$ and $q(x)$, which we label as L and M respectively. Since these are finite degree extensions whose intersection is equal to $\mathbb{Q}$, it follows that

$$\operatorname{Gal}(K/\mathbb{Q}) \cong \operatorname{Gal}(L/\mathbb{Q}) \times \operatorname{Gal}(M/\mathbb{Q}).$$

We start by seeing that $L = \mathbb{Q}\left(\omega, \sqrt[3]{11}\right)$. Since $p(x)$ has degree 3 and not all of its roots are real, it follows that $\operatorname{Gal}(L/\mathbb{Q}) \cong S_3$, while since M has degree 2 over $\mathbb{Q}$, it follows that $\operatorname{Gal}(M/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z}$, so that $\operatorname{Gal}(K/\mathbb{Q}) \cong S_3 \times \mathbb{Z}/2\mathbb{Z}$.

Problem (January '98, Problem 5)**:** The Galois group G of a polynomial $f(x) \in F[x]$ of degree 4 is known to contain a subgroup isomorphic to the dihedral group D_4.

(a) Show that $f(x)$ is irreducible.

(b) If some root r_i of $f(x)$ lies in the subfield $F\left(r_j, r_k\right)$ generated by two other roots, show that $F\left(r_j, r_k\right)$ is a splitting field for $f(x)$ and that $G = D_4$.

Solution:

(a) We observe that the group S_4 is of order 24, and by the Sylow theorems, there are three conjugate subgroups of order 8, each of which is isomorphic to D_4. Since each of these copies of D_4 is transitive, it follows that the Galois group of $f(x)$ acts transitively on the roots of $f(x)$. Since the Galois group of $f(x)$ is transitive, it follows that $f(x)$ is irreducible.

(b)

Problem (Algebra II 2019 Midterm II, Problem 2): Let K/F be a finite Galois extension with $G = \mathrm{Gal}(K/F)$.

(a) Assume that G is a simple group, with $\alpha \in K \setminus F$ and $\mu_{\alpha,F}(x)$ the minimal polynomial of α over F. Prove that K is a splitting field for $\mu_{\alpha,F}(x)$.

(b) Let $n = [K : F]$, and fix integers m and ℓ with $m\ell = n$. Find a condition on the subfield lattice of K/F which is equivalent to the following: G can be written as a semidirect product $A \rtimes B$ for some subgroups A and B with $|A| = m$ and $|B| = \ell$.

Solution:

(a)

(b) By the definition of the semidirect product, we have that $A \trianglelefteq G$, $G = AB$, and $A \cap B = \{e\}$. In particular, since $|AB| = \frac{|A||B|}{|A\cap B|}$, we have that $G = AB$ is equivalent to $A \cap B = \{e\}$.

We have that $A \trianglelefteq G$ if and only if K^A/F is Galois, that $G = AB$ if and only if $K^{AB} = F$, and that $A \cap B = \{e\}$ if and only if $K^{A\cap B} = K$.

Observe now that $K^{AB} = K^A \cap K^B$. This follows from the fact that $\alpha \in K^{AB} = F$ if and only if $g(\alpha) = \alpha$ for any $g \in AB$, hence for all $g \in A$ and $g \in B$.

Therefore, we claim that $G = A \rtimes B$ with $|A| = m$ and $|B| = \ell$ if and only if K/F contains subfields M_A and M_B such that

(i) $[M_A : F] = \ell$, $[M_B : F] = m$;

(ii) M_A/F is Galois;

(iii) $M_A \cap M_B = F$.

The forward direction follows from our earlier work by setting $M_A = K^A$ and $M_B = K^B$.

Problem (DF Problem 14.2.12): Determine the Galois group of the splitting field over $\mathbb{Q}$ of $x^4 - 14x^2 + 9$.

Solution: Factorizing over $\mathbb{C}$, we observe that

$$\begin{aligned}
p(x) &= x^4 - 14x^2 + 9\\
&= \left(x^2 - 7\right)^2 - 40\\
&= \left(x - \sqrt{7+2\sqrt{10}}\right)\left(x + \sqrt{7+2\sqrt{10}}\right)\left(x - \sqrt{7-2\sqrt{10}}\right)\left(x + \sqrt{7-2\sqrt{10}}\right).
\end{aligned}$$

If we let

$$\begin{aligned}
\alpha &= \sqrt{7+2\sqrt{10}}\\
\beta &= -\sqrt{7+2\sqrt{10}}\\
\gamma &= \sqrt{7-2\sqrt{10}}\\
\delta &= -\sqrt{7-2\sqrt{10}},
\end{aligned}$$

then we observe that $\alpha\gamma = 3$, whence $\gamma \in \mathbb{Q}(\alpha)$ so that $\mathbb{Q}(\alpha)/\mathbb{Q}$ is the splitting field for $p(x)$. Since $p(x)$ is irreducible, it follows that $\mathbb{Q}(\alpha)/\mathbb{Q}$ has degree 4. Furthermore, since $p(x)$ is a separable irreducible polynomial, there are automorphisms taking α to every other root, giving the automorphism group

$$G = \{\mathrm{id}, (\alpha \mapsto \gamma), (\alpha \mapsto \beta), (\alpha \mapsto \delta)\}.$$

Observe that all of these automorphisms fix $\mathbb{Q}$ and have order 2, meaning that $G \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Problem (August '21, Problem 6): Consider $f(x) = x^4 + ax^2 - 1 \in \mathbb{Q}[x]$, where $a \in \mathbb{Z} \setminus \{0\}$. Let K be the splitting field of $f(x)$ over $\mathbb{Q}$.

(a) Show that $f(x)$ is irreducible.

(b) Show that $i \in K$.

(c) Determine the isomorphism class of the Galois group $\operatorname{Gal}(K/\mathbb{Q})$.

Solution:

(a) From the rational roots theorem, we know that any rational root of $f(x)$ must be either 1 or -1. Since both $f(1)$ and $f(-1)$ are equal to a, and $a \neq 0$, it follows that $f(x)$ has no rational roots, so $f(x)$ is irreducible.

(b) Since $f(x)$ is irreducible over K, it follows that $f(x)$ is separable, and upon completing the square we may write

$$x^2 + ax^2 - 1 = (x-\alpha)(x+\alpha)(x-\beta)(x+\beta)$$

for some $\alpha, \beta \in K$, meaning that we have $\alpha^2\beta^2 = -1$. Therefore, it follows that $\alpha\beta = i$, so $i \in K$.

(c) To determine the isomorphism class of the Galois group $\operatorname{Gal}(K/\mathbb{Q})$, we start by considering the three roots

$$\begin{aligned}
\theta_1 &= 0\\
\theta_2 &= -(\alpha+\beta)^2\\
&= -\left(\alpha^2 + \beta^2 + 2i\right)\\
\theta_3 &= -(\alpha-\beta)^2\\
&= -\left(\alpha^2 + \beta^2 - 2i\right)
\end{aligned}$$

Observe that $\theta_1 \in \mathbb{Q}$, meaning that the Galois group is isomorphic to either D_4 or $\mathbb{Z}/4\mathbb{Z}$ as θ_1 is stabilized by the Galois group. To distinguish, we will compute the order of $K = \mathbb{Q}(\alpha, i)$.

For this, it suffices to compute $[K : \mathbb{Q}(\alpha)]$. Since we may write

$$\begin{aligned}
\alpha &= \sqrt{\frac{-a + \sqrt{1+a^2}}{2}}\\
&\in \mathbb{R},
\end{aligned}$$

it follows that $\mathbb{Q}(\alpha) \subsetneq K$, so that

$$\begin{aligned}
[K : \mathbb{Q}] &= [K : \mathbb{Q}(\alpha)][\mathbb{Q}(\alpha) : \mathbb{Q}]\\
&= 8,
\end{aligned}$$

where we use the fact that f is irreducible to determine that $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$.

Problem (August '15, Problem 8): Consider the polynomial $x^6 - 3$ over $\mathbb{Q}$. What is the degree of its splitting field, and what is the splitting field? Describe the Galois group of the splitting field as a subgroup of the symmetric group S_6. Is it abelian?

Solution: We start by observing that $x^6 - 3$ is a polynomial in $\mathbb{Z}[x]$ in addition to $\mathbb{Q}[x]$, so we may use the Eisenstein criterion to see that it is irreducible as 3 is prime, so by Gauss's Lemma, it is irreducible over $\mathbb{Q}$.

If ω denotes a primitive sixth root of unity, then we observe that the roots of the polynomial are given by

$$A = \left\{\sqrt[6]{3}, \omega\sqrt[6]{3}, \omega^2\sqrt[6]{3}, \omega^3\sqrt[6]{3}, \omega^4\omega\sqrt[6]{3}, \omega^5\sqrt[6]{3}\right\}.$$

We start by finding the degree of the splitting field K over $\mathbb{Q}$. For this, observe that $K \subseteq \mathbb{Q}\left(\omega, \sqrt[6]{3}\right)$, while $\omega = \frac{\omega\sqrt[6]{3}}{\sqrt[6]{3}}$, whence $K = \mathbb{Q}\left(\omega, \sqrt[6]{3}\right)$.

Since the 6th cyclotomic polynomial is given by $x^2 - x + 1$, and the minimal polynomial for $\sqrt[6]{3}$ is $x^6 - 3$, we have

$$\begin{aligned}[K : \mathbb{Q}] &= \left[K : \mathbb{Q}\left(\sqrt[6]{3}\right)\right]\left[\mathbb{Q}\left(\sqrt[6]{3}\right) : \mathbb{Q}\right]\\ &= (2)(6)\\ &= 12,\end{aligned}$$

where we use the fact that $\sqrt[6]{3} \in \mathbb{R}$ to find that $\left[K : \mathbb{Q}\left(\sqrt[6]{3}\right)\right] = 2$. Since the polynomial $x^6 - 3$ is irreducible and $\mathbb{Q}$ is characteristic zero, it follows that the Galois group is transitive in its action on A. Considering the two $\mathbb{Q}$-automorphisms given by

$$\begin{aligned}\sigma &= \begin{cases}\sqrt[3]{6} \mapsto \omega\sqrt[3]{6}\\ \omega \mapsto \omega\end{cases}\\ \tau &= \begin{cases}\sqrt[3]{6} \mapsto \sqrt[3]{6}\\ \omega \mapsto \omega^{-1},\end{cases}\end{aligned}$$

we observe that σ corresponds to the element of S_6 given by the 6-cycle

$$\sigma' = (1, 2, 3, 4, 5, 6),$$

while τ corresponds to the product of transpositions given by

$$\tau' = (2, 6)(3, 5).$$

> **Remark:** Observe that σ and τ are well-defined $\mathbb{Q}$-automorphisms because $\omega\sqrt[3]{6}$ is another root of the minimal polynomial $x^6 - 3$, while τ is the restriction of complex conjugation to K.

Additionally, we see that σ and τ satisfy the relations

$$\begin{aligned}\sigma^6 &= \mathrm{id}\\ \tau^2 &= \mathrm{id}\\ \tau\sigma &= \sigma^{-1}\tau,\end{aligned}$$

so that the group is non-abelian. Observe that these are relations for any homomorphic image of the dihedral group D_6 of order 12, while σ and τ must generate a group of order at least 12 since no power of σ fixes two elements of A. Therefore, it follows that the Galois group is isomorphic to D_6, which is non-abelian.

Problem (August '01, Problem 8): Find the Galois group of $f(x) = x^{13} - 1$ over $\mathbb{Q}$.

Solution: Observe that 13 is prime, so

$$x^{13} - 1 = (x - 1)\Phi_{13}(x),$$

meaning that the Galois group of $f(x)$ is equal to the Galois group of the cyclotomic polynomial $\Phi_{13}(x)$. The Galois group of the cyclotomic polynomial is given by $(\mathbb{Z}/13\mathbb{Z})^{\times} \cong \mathbb{Z}/12\mathbb{Z}$ by the characterization of Galois groups of cyclotomic polynomials.

Problem (January '19, Problem 7): Let $F = \mathbb{Q}\left(\sqrt[6]{3}, i\right)$.

(a) Prove that $[F : \mathbb{Q}] = 12$.

(b) Prove that the extension $F/\mathbb{Q}$ is Galois.

(c) Prove that the Galois group $\operatorname{Gal}(F/\mathbb{Q})$ is isomorphic to D_6, the dihedral group of order 12.

Solution:

(a) We start by observing that we may write

$$[F : \mathbb{Q}] = \left[F : \mathbb{Q}\left(\sqrt[6]{3}\right)\right]\left[\mathbb{Q}\left(\sqrt[6]{3}\right) : \mathbb{Q}\right].$$

The minimal polynomial of $\sqrt[6]{3}$ is given by $\mu(x) = x^6 - 3$, whose irreducibility follows from the Eisenstein criterion. In particular, this means that $\left[\mathbb{Q}\left(\sqrt[6]{3}\right) : \mathbb{Q}\right] = 6$. Furthermore, note that $\mathbb{Q}\left(\sqrt[6]{3}\right) \subseteq \mathbb{R}$, so since $F \not\subseteq \mathbb{R}$, it follows that $\left[F : \mathbb{Q}\left(\sqrt[6]{3}\right)\right] = 2$, so that $[F : \mathbb{Q}] = 12$.

(b) We claim that F is the splitting field for $\mu(x) = x^6 - 3$. For this, observe that, since $\sqrt[6]{3} \in F$, so too is $\sqrt{3}$. Observe that the splitting field K of $\mu(x)$ can be given by $\mathbb{Q}\left(\sqrt[6]{3}, \zeta_6\right)$, where ζ_6 is a primitive sixth root of unity. One such primitive sixth root of unity can be written as

$$\zeta_6 = \frac{1}{2} + \frac{\sqrt{3}}{2} i,$$

so that $F = K$. Since F is the splitting field of a separable polynomial (since $\mathbb{Q}$ is characteristic zero), it follows that $F/\mathbb{Q}$ is Galois.

(c) We may consider the roots of $\mu(x) = x^6 - 3$ as

$$A = \left\{\sqrt[6]{3}, \zeta_6\sqrt[6]{3}, \zeta_6^2\sqrt[6]{3}, \zeta_6^3\sqrt[6]{3}, \zeta_6^4\sqrt[6]{3}, \zeta_6^5\sqrt[6]{3}\right\},$$

labeled 1 through 6 respectively. Since $x^6 - 3$ is irreducible and separable, the Galois group is transitive. Considering the maps

$$\sigma = \begin{cases} \sqrt[6]{3} \mapsto \zeta_6\sqrt[6]{3} \\ \zeta_6 \mapsto \zeta_6 \end{cases}$$

$$\tau = \begin{cases} \sqrt[6]{3} \mapsto \sqrt[6]{3} \\ \zeta_6 \mapsto \zeta_6^{-1}, \end{cases}$$

we observe that these correspond to the permutations in S_6 given by

$$\begin{aligned} \sigma' &= (1, 2, 3, 4, 5, 6) \\ \tau' &= (2, 6)(3, 5). \end{aligned}$$

These permutations satisfy the relations

$$\begin{aligned} \sigma^6 &= \mathrm{id} \\ \tau^2 &= \mathrm{id} \\ \tau\sigma\tau &= \sigma^{-1}, \end{aligned}$$

where the first two emerge from the cycle decomposition and the third emerges from the properties of conjugation in the symmetric group. That σ and τ are $\mathbb{Q}$-automorphisms emerges from the fact that the first map does not affect any elements of $\mathbb{Q}$ while the second map is complex conjugation restricted to F. This gives that the group $\langle \sigma, \tau \rangle$ is a quotient of D_6. Furthermore, since no power of σ fixes any element, it must be the case that $|\langle \sigma, \tau \rangle| \geq 12$, so that the group generated by σ and τ is isomorphic to D_6.

Problem (DF 14.2.17 (d))**:** Let K/F be any finite extension, and let $\alpha \in K$. Let L be a Galois extension of F containing K, and let $H \leq \operatorname{Gal}(L/F)$ be the subgroup corresponding to K. Define the *norm* of α by

$$N_{K/F}(\alpha) = \prod_{\sigma} \sigma(\alpha),$$

where the product is taken over all embeddings of K into an algebraic closure of F — equivalently, a set of coset representatives of H in $\operatorname{Gal}(L/F)$. Note that if K/F is Galois, then this is given b y

$$N_{K/F}(\alpha) = \prod_{\sigma \in \operatorname{Gal}(K/F)} \sigma(\alpha).$$

Suppose

$$m_\alpha(x) = x^d + a_{d-1}x^{d-1} + \cdots + a_1 x + a_0$$

is the minimal polynomial for $\alpha \in K$ over F. Let $n = [K : F]$. Prove that d divides n, there are d distinct Galois conjugates of α over that are repeated n/d times in the product above, and conclude that

$$N_{K/F}(\alpha) = (-1)^n a_0^{n/d}.$$

Solution: We start by observing that L/F is separable, meaning m_α is separable, so there are exactly d Galois conjugates $\alpha_1, \dots, \alpha_d$ such that $\alpha_1 \cdots \alpha_d = (-1)^d a_0$. Furthermore, observe that

$$\begin{aligned} n &= [K : F] \\ &= [K : F(\alpha_1, \dots, \alpha_d)][F(\alpha_1, \dots, \alpha_d) : F] \\ &= md, \end{aligned}$$

so that d divides n.

Since the number of embeddings of K into an algebraic closure of F is given by $[K : F] = [G : H]$, it follows that the product is taken over n iterations, so by Vieta's formulas we have

$$\begin{aligned} N_{K/F}(\alpha) &= \prod_{\sigma} \sigma(\alpha) \\ &= \left(\prod_{\theta} \theta(\alpha)\right)^{n/d} \\ &= \left((-1)^d a_0\right)^{n/d} \\ &= (-1)^n a_0^{n/d}. \end{aligned}$$

Problem (DF 14.2.18 (d))**:** Define the trace of α from K to F by

$$\operatorname{Tr}_{K/F}(\alpha) = \sum_{\sigma} \sigma(\alpha)$$

a sum of Galois conjugates of α where the sum is over all embeddings of K into a fixed algebraic closure of F. Prove that

$$\operatorname{Tr}_{K/F}(\alpha) = -\frac{n}{d} a_{d-1}.$$

Solution: Observe that by our previous work, the total sum is given by $[K : F] = n$, so by Vieta's formulas, we have

$$\operatorname{Tr}_{K/F}(\alpha) = \sum_{\sigma} \sigma(\alpha)$$

$$\begin{aligned}&= \frac{n}{d}\left(\sum_{\theta}\theta(\alpha)\right)\\&= -\frac{n}{d}a_{d-1}.\end{aligned}$$